FLOOR PLANNING

FP PLAN

Floor Planning creates the blueprint of the chip by arranging major blocks, planning I/O, power and ensuring an optimized area, performance and routability.

From Concept to Layout — It Starts with a Good Floor Plan.

INPUT FROM SYNTHESIS
- Gate Level Netlist
- Libraries
- Constraints (SDC)

DESIGN UNDERSTANDING
- Block Analysis
- I/O Analysis

MACRO PLANNING
- Macro Selection
- Macro Placement

FLOORPLAN CREATION
- Die & Core Area
- I/O Placement
- Power Plan

PLACEMENT
- Macro Placement
- Blockage / Keepout
- Channel Planning

CHECKS & VALIDATION
- DRC / LVS
- Antenna
- Connectivity

READY FOR NEXT STAGE

DESIGN ANALYSIS
- Block Diagram
- Hierarchy
- Area Report

MACRO IDENTIFICATION
- Hard Macros
- SRAMs / IPs
- Analog Blocks

DIE & CORE DEFINITION
- Die Area
- Core Area
- Aspect Ratio

POWER PLAN
- Power Straps
- RDL / Rings
- Well Taps

SIGNOFF

DIE & CORE AREA

I/O PLACEMENT

POWER PLANNING

BLOCKAGES & KEEP OUTS

VALIDATION & SIGNOFF

FLOOR PLANNING FLOW

1. DIE & CORE DEFINITION
   - Die Size
   - Core Size
   - Aspect Ratio
2. I/O PLACEMENT
   - Pad Placement
   - Corner / Edge
   - I/O Balance
3. MACRO PLACEMENT
   - Macro Positioning
   - Orientation
   - Channel Planning
4. POWER PLANNING
   - Power Rings
   - Power Straps
   - Well Taps
5. CHECKS & VALIDATION
   - DRC / LVS
   - Antenna Check
   - Connectivity

FLOOR PLANNING GOALS

- Utilize Area Efficiently
- Enable High Performance
- Improve Routability
- Ensure Power Integrity
- Thermal Management
- Timing Closure
- DFM Compliance

KEY ENGINES IN FLOOR PLANNING

AREA ENGINE
- Area Estimation
- Utilization
- Density Check

I/O ENGINE
- Pad Ringing
- I/O Distribution
- Signal Grouping

PLACEMENT ENGINE
- Macro Placement
- Legalization
- Optimization

POWER ENGINE
- IR Drop Analysis
- EM Analysis
- Power Grid Opt.

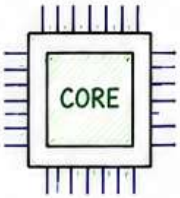
TIMING ENGINE
- Timing Estimation
- Critical Paths
- Slack Analysis

DFM ENGINE
- DRC Clean
- Antenna Clean
- Manufacturable

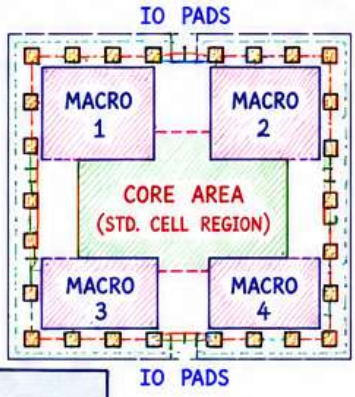
A Good Floor Plan is the Foundation of a Successful Chip.
≥ Plan Smart. Route Easy. Deliver Silicon. ≤

Author :
Konda Tharun Vivekananda

INPUT FROM SYNTHESIS → DESIGN ANALYSIS → MACRO PLACEMENT → POWER PLANNING → I/O PLACEMENT → VALIDATION & SIGNOFF → READY FOR NEXT STAGE



FLOOR PLANNING INDEX



FLOORPLAN LEGEND

- CORE AREA
- MACROS
- IO PADS
- === POWER RING
- --- ROUTING CHANNEL
- DIE BOUNDARY

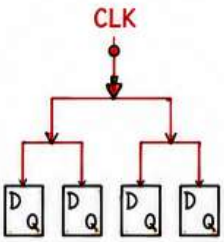
KEY GOALS

- Timing
- Power
- Area
- Congestion
- IR Drop
- Reliability

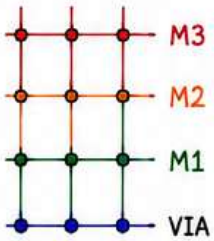


S.No.	Topic
1	Floor RTL-to-Synthesis and Complete Floorplan Flow
2	Floor Planning – Introduction
3	Floor Plan Important Terms
4	Floor Planning Steps in ICC2 Tool
5	Macro and Macro Placement
6	Halo (Keep-Out) and Pin Planning
7	Flywire Analysis
8	IO and Pin Planning
9	Power Planning
10	Special Cells During Floorplan
11	Congestion Planning in Floorplan
12	Blockages in Floorplan
13	Floorplanning Optimization
14	Floorplanning – Problems and Their Solutions
15	Important Terms of Floorplanning
16	Flat Fullchip and Hierarchical Fullchip Floorplan
17	Multi Voltage Domain Floorplanning
18	Domain Placement Strategy
19	Isolation Cells
20	Retention Flops
21	AON Clamps on Domains
22	Power Grid Separation and Power Switch
23	Multi Voltage Verification

CLOCK TREE



POWER GRID

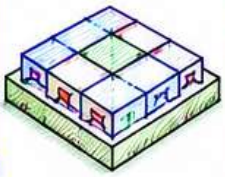


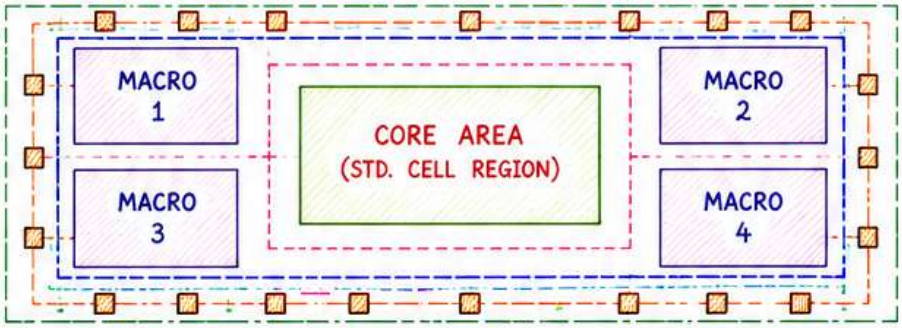
DESIGN RULES

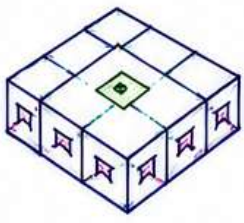
- ✓ Spacing
- ✓ Width
- ✓ Enclosure
- ✓ Density

KEEP-OUT ZONE









PD FLOOR PLAN

↓ = Flow
□ = Stage
✱ = Key

DEFINITION :

Floor planning is the first step in Physical Design. It decides "WHERE" the main blocks of the chip will be placed on the die and how they will communicate. A good floorplan is the foundation for timing, power, routability and reliability.

1. FROM RTL TO SYNTHESIS AND THEN TO FLOOR PLAN – FLOW

SYSTEM SPECIFICATION
↓
ARCHITECTURE DESIGN
↓
RTL DESIGN
↓
TESTBENCH + FUNCTIONAL VERIFICATION
↓
VERIFIED RTL
↓
SYNTHESIS INPUTS
(RTL + .lib + .lef + .sdc + UPF + TECHNOLOGY FILES)
↓
SYNTHESIS
↓
ELABORATION
↓
SDC READ + SANITY CHECKS
↓
GENERIC MAPPING
↓
TECHNOLOGY MAPPING
↓
LOGIC / TIMING / AREA / POWER OPTIMIZATION
↓
DFT / SCAN INSERTION
↓
POST-DFT OPTIMIZATION
↓
GATE-LEVEL NETLIST
↓
SYNTHESIS SIGNOFF / QOR CHECKS
↓
- - - - - - → HANDOFF TO PHYSICAL DESIGN - - - - - -
↓
FLOORPLAN INITIALIZATION
↓
DIE & CORE AREA DEFINITION
↓
IO / PIN PLACEMENT
↓
MACRO PLACEMENT
↓
POWER PLANNING
↓
STANDARD CELL PLACEMENT
↓
✱ FLOORPLAN ✱

✱ FLOW EXPLANATION :

- RTL Design creates the digital description of the chip.
- Synthesis converts RTL into Gate-level netlist using libraries and constraints.
- Outputs after signoff are handed over to Physical Design.
- Floorplan decides the placement of major blocks, I/O, power grid and provides the base for placement and routing.

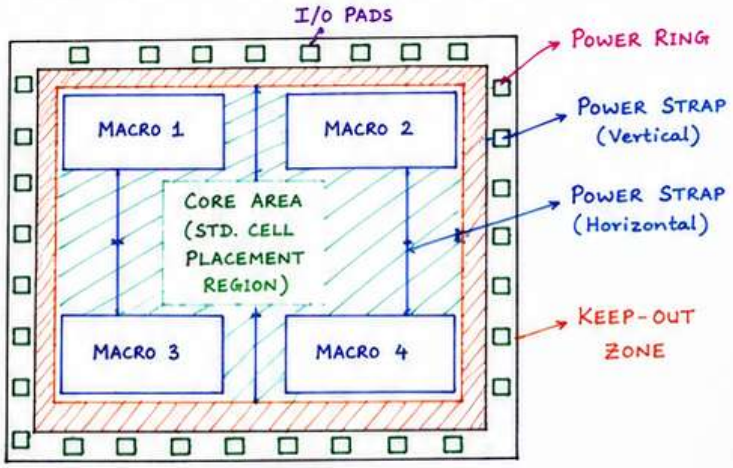
✱ FLOORPLAN GOALS :

- Meet timing, area, power and DRC targets.
- Good routability and congestion free design.
- Balanced wirelength and performance.
- Proper power distribution and IR drop control.
- Thermal and reliability friendly layout.

✱ FLOORPLAN MAIN COMPONENTS :

1) DIE AREA → Total chip area including pads.
2) CORE AREA → Area available for standard cells.
3) I/O PADS → Interface to external world.
4) MACROS / IP → Large blocks (RAM, PLL, CPU, etc.).
5) POWER NETWORK → Power rings, straps, meshes.
6) PLACEMENT REGION → Area where Std. cells will be placed.
7) KEEP-OUT ZONES → Reserved areas for design rules.

✱ BASIC FLOORPLAN VIEW (TOP VIEW) :



NOTE :

- Floorplan is done before placement.
- Changes in floorplan later may cause timing, congestion and IR drop issues.
- A good floorplan saves > 50% of design effort.

AUTHOR :
KONDA THARUN VIVEKANANDA

PD FLOOR PLANNING

2. FLOOR PLANNING – INTRODUCTION

- Floor planning is the first step in Physical Design after synthesis. It decides the physical organization of major blocks on the die.
- It defines die size, core area, IO locations, macro placement, power grid and routing resources.
- A good floorplan is the foundation for timing, power, routability, area and reliability.

WHY WE DO FLOOR PLANNING ?

- Defines physical boundaries of the design.
- Places macros, IOs and keeps keep-out / no-place zones.
- Allocates routing channels to avoid congestion.
- Ensures proper power distribution and IR drop control.
- Improves timing closure by reducing interconnect length.
- Helps in meeting DRC, reliability and manufacturing rules.
- Provides a stable base for placement and CTS.

ISSUES OCCUR FOR INCORRECT FLOORPLAN

- Congestion in routing → detours → timing degradation.
- Long interconnects → high delay and high power.
- IR drop violations due to poor power grid placement.
- Placement failures or poor utilization of area.
- Clock skew increases due to improper block locations.
- DRC violations and routing rule violations.
- More ECOs, re-spins and increased design cost.

HOW TO AVOID FLOORPLAN ISSUES ?

- Understand design requirements and constraints clearly.
- Identify critical macros and place them early.
- Keep high switching blocks near power sources.
- Allocate routing channels : horizontal & vertical.
- Maintain sufficient spacing between macros.
- Plan power rings and stripes from the beginning.
- Keep IOs on all four sides for better connectivity.
- Run early congestion and IR drop estimation.
- Follow physical design guidelines and rule decks.
- Validate floorplan with DRC, connectivity and PPA checks.

* KEY POINT *

Floorplanning decides WHERE everything will be placed and HOW they will communicate.
A good floorplan saves more than 50% of the design effort in later stages.

PRACTICAL OBSERVATION

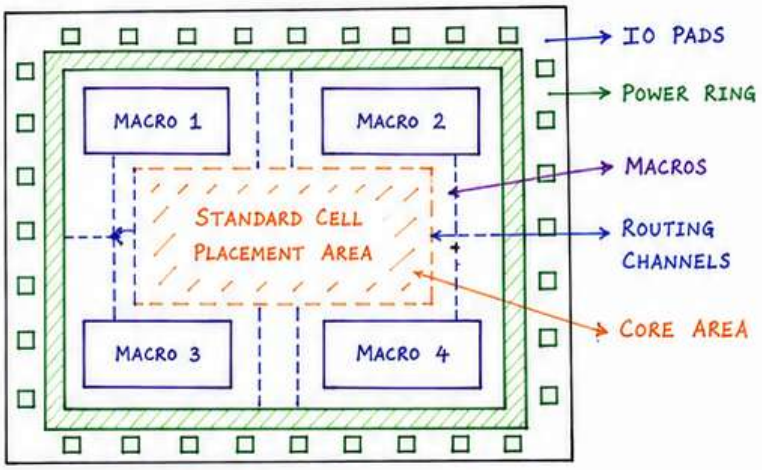
- Poor floorplan → congestion ↑ → timing ↓ → more ECOs.
- Proper macro placement and power planning reduce IR drop and improve timing closure.
- Early verification of floorplan reduces risk in later steps.

VIVA QUESTIONS

Q1. Why is floor planning important in VLSI Physical Design ?
→ It defines the physical organization of the chip and directly impacts timing, power, area and routability.

Q2. What are the main outputs of floor planning ?
→ Die & core areas, IO locations, macro placement, power plan, keep-out zones and floorplan DEF.

SIMPLE FLOORPLAN VIEW :

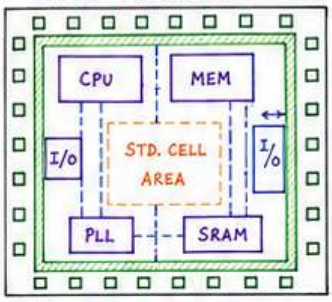


FLOORPLAN COMPONENTS :

COMPONENT		DESCRIPTION
DIE AREA	:	Total chip area including pads and scribe.
CORE AREA	:	Area available for standard cells and routing.
IO PADS	:	Interface points for external connections.
MACROS / IP	:	Large blocks like RAM, PLL, CPU, etc.
POWER RING	:	Surrounds core to distribute power.
POWER STRIPES	:	Horizontal / vertical wires for power grid.
KEEP-OUT ZONE	:	Reserved area for routing or blockage.
PLACEMENT REGION	:	Area where standard cells can be placed.
ROUTING CHANNEL	:	Space reserved for signal routing.

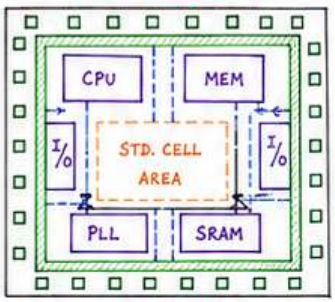
GOOD vs BAD FLOORPLAN :

GOOD FLOORPLAN



- ✓ Short interconnects
- ✓ Balanced block distribution
- ✓ Low congestion
- ✓ Better timing closure
- ✓ Less power (IR drop)
- ✓ High routability

BAD FLOORPLAN



- ✗ Long interconnects
- ✗ Macros not balanced
- ✗ High congestion
- ✗ Poor timing
- ✗ High IR drop
- ✗ Routing failures

FLOORPLANNING CHECKLIST :

- ☑ Die & core area defined
- ☑ IO pads placed
- ☑ Macros placed
- ☑ Power ring defined
- ☑ Power stripes planned
- ☑ Routing channels allocated
- ☑ Keep-out zones set
- ☑ Physical rules checked
- ☑ Congestion estimation done
- ☑ Floorplan DEF generated

SUMMARY POINTS

- Floorplanning is the foundation of Physical Design.
- It decides die, core, macros, IOs, power and routing.
- Good floorplan → less congestion → better timing.
- Poor floorplan → delay, high power, re-work, cost ↑.
- Early planning saves time, cost and improves QoR.

AUTHOR : Konda Tharun Vivekananda

PD FLOOR PLANNING

3. Floor Plan Important Terms

1) Core Area

- The usable area inside the die boundary where standard cells are placed.
- Core area excludes IO pads, power ring, power stripes and other blockages.

$$\text{Core Area} = \text{Core.Width} \times \text{Core.Height}$$

2) Die Area

- Total silicon area of the chip including everything (core + IO + macros + rings).
- Die is the outermost boundary of the chip.

$$\text{Die Area} = \text{Die.Width} \times \text{Die.Height}$$

3) Aspect Ratio (AR)

- Ratio between the longer side and shorter side of the core.
- Controls routability and placement quality.

$$AR = \frac{\text{Longer Side}}{\text{Shorter Side}} \qquad AR \geq 1$$

4) Utilization (U)

- Percentage of core area actually used by standard cells.
- It should be neither too high nor too low.

$$U(\%) = \frac{\text{Total Cell Area}}{\text{Core Area}} \times 100$$

Why We Need Good Floorplan ?

- Provides a good foundation for timing, power & routability.
- Reduces interconnect length and congestion.
- Improves timing closure and reduces skew.
- Ensures proper power distribution and IR drop control.
- Minimizes ECOs and design re-spins.
- Helps in meeting PPA (Power, Performance, Area) goals.

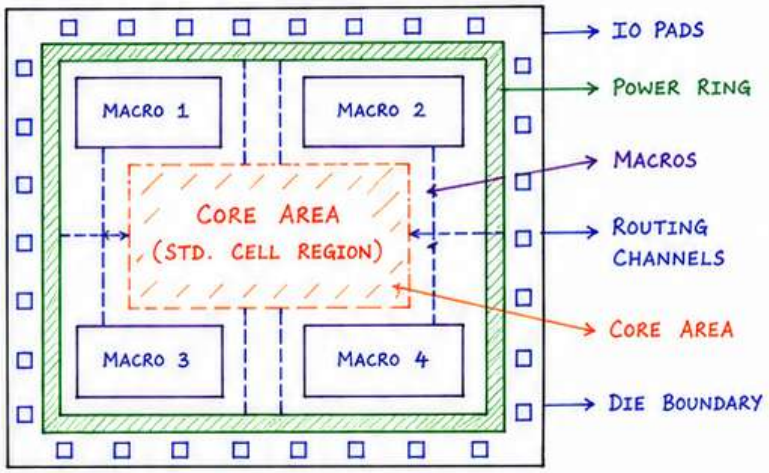
Issues If Floorplan Is Incorrect

1. High congestion → Detours → Timing degradation
2. Long interconnects → High delay & power
3. Poor power network → IR drop → Timing fails
4. Low utilization → Wastage of silicon area
5. Critical paths cut by blockages → More buffers
6. Poor IO placement → Routing congestion
7. Unbalanced macro placement → Heat / reliability

* Key Point *

A good floorplan decides the success of the entire Physical Design. Most of the problems in later stages come from a bad floorplan.

Typical Floorplan (Top View)



Formulas Summary

- Core Area = $\text{Core}_W \times \text{Core}_H$
- Die Area = $\text{Die}_W \times \text{Die}_H$
- Aspect Ratio = $\frac{\text{Longer Side}}{\text{Shorter Side}}$ (≥ 1)
- Utilization (%) = $\left(\frac{\text{Total Cell Area}}{\text{Core Area}}\right) \times 100$
- Core to Die Util. = $\left(\frac{\text{Core Area}}{\text{Die Area}}\right) \times 100$
- Core Margin (%) = $\left(\frac{\text{Core Area}}{\text{Die Area}}\right) \times 100$

How To Solve Floorplan Issues

1. Choose proper core size → Leave enough margin.
2. Maintain AR in recommended range (1 to 2 preferred).
3. Place macros early and group related macros.
4. Provide proper routing channels (H & V).
5. Plan power rings & stripes from the beginning.
6. Place IOs close to macros and critical blocks.
7. Run early congestion and IR drop estimation.
8. Follow design rules and physical guidelines.

Additional Note Points

A) Fixed Outline Floorplan

- Die outline is fixed by technology / package / customer.
- We can change only core size and internal placement.
- Must fit design within fixed die.

B) Rectilinear Floorplan

- All the shapes are made of rectangles and straight edges.
- Easy to implement and tool friendly.
- Recommended for most of the ASIC designs.

Practical Observation

* Poor floorplan → more ECOs and timing problems.
* Start with rough floorplan → refine iteratively.
* Keep congestion and IR drop in check early.
* Floorplan is cheap to fix in start, very costly later.

Viva Questions

Q1. What is core area ?
→ Usable area inside the die where standard cells are placed.

Q2. What is utilization ?
→ Percentage of core area occupied by standard cells.

Q3. Why aspect ratio is important ?
→ It affects routability, wirelength and timing.

Floorplan Checklist

- ☑ Core & die area defined
- ☑ Aspect ratio in range
- ☑ Macros placed
- ☑ IOs placed on all sides
- ☑ Power rings and stripes
- ☑ Routing channels created
- ☑ Utilization in target range

Summary Points

- Core area is the usable region for placement.
- Die area is the total chip area.
- Good aspect ratio improves routability.
- Utilization should be balanced.
- Incorrect floorplan causes congestion, IR drop and timing failures.
- Plan early, verify early, close early.

AUTHOR : Konda Tharun Vivekananda

PD FLOOR PLANNING

4. Floor Planning Steps In ICC2 Tool (Step by Step)

Floorplanning in ICC2 establishes the physical framework of the design before placement. Below are the typical step-by-step commands and actions.

① Create Floorplan (Initialize)

- Initialize design and create design boundary.

```
create_floorplan -core_utilization <0.6>
        -die_area {0 0 <dieW> <dieH>}
```

Output: Die & Core created

② Set Floorplan Mode

- Enter floorplan mode.

```
set_floorplan_mode -plan
```

Output: Ready for floorplan

③ Read Tech And Design Data

- Load LEF/DEF and netlist to get design info.

```
read_lef <tech.lef>
read_def <design.def> or read_db <netlist.db>
```

Output: Design data loaded

④ Define Die Area

- Define or modify die size.

```
create_die_area -die_area {0 0 <dieW> <dieH>}
```

Output: Die area defined

⑤ Define Core Area

- Create core area inside die with margins.

```
create_core_area -core_area {m m [expr <dieW>-m]
                    [expr <dieH>-m]}
```

Output: Core area defined

⑥ Place IO Pins

- Place IOs on all four sides.

```
create_io_placement -side {top bottom left right}
```

Output: IOs placed

⑦ Create Power Ring / Stripes (Optional at FP)

- Create power ring around core and power stripes.

```
create_power_ring -nets {VDD VSS}
create_power_stripe -layer M4 -direction vertical
```

Output: Power grid created

⑧ Place Macros / Blocks

- Place large macros and hard IPs.

```
place_blockages -auto_place
        or
place_instance <inst_name> <x> <y> <orient>
```

Output: Macros placed

⑨ Define Rows For Standard Cells

- Define placement rows inside core.

```
create_placement -core_rows
```

Output: Rows created

⑩ Create Placement Blockages (Keep-out)

- Blockages for macros / power / routing channels.

```
create_placement_blockage -type hard|soft
        -rect {x1 y1 x2 y2}
```

Output: Blockages defined

⑪ Save Floorplan

- Save the floorplan database.

```
save_mw_cel -as floorplan_init
```

Output: Floorplan saved

⑫ Check Floorplan

- Run basic DRC and floorplan checks.

```
check_floorplan
```

Output: Reports

Practical Order Flow

① → ② → ③ → ④ → ⑤ → ⑥
⑦ → ⑧ → ⑨ → ⑩ → ⑪ → ⑫

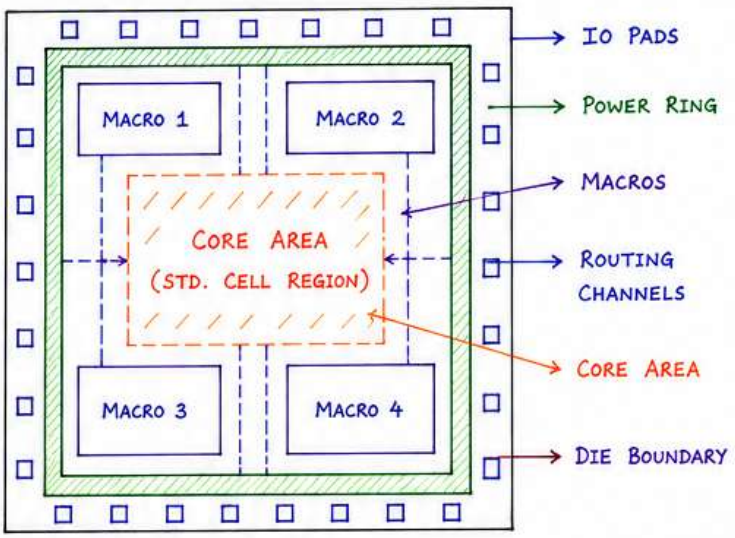
Important Notes

- All dimensions in database units (usually microns).
- Core area should exclude all non-standard cell regions.
- Keep proper margins for ESD, seal ring and dummies.
- Follow technology / foundry design rules.

*** Key Point ***

A good floorplan reduces congestion, improves timing and saves area, power and design effort in later stages.

Typical Floorplan (Top View)



Floorplan Views In ICC2

1) Top View : To see overall die, core, macros, IOs.
2) Power View : To view power rings, stripes and rails.
3) Placement View : To view rows, blockages and channels.

Floorplan Checklist

- ☑ Die area and core area defined
- ☑ Core utilization in target range
- ☑ Aspect ratio in recommended range (1 to 2)
- ☑ IO pins placed on all four sides
- ☑ Macros placed with proper spacing
- ☑ Power rings and stripes created
- ☑ Routing channels (H & V) available
- ☑ Placement rows created
- ☑ Keep-out / blockages defined
- ☑ Floorplan DRC clean

Recommended Utilization

ASIC : 50% - 70%
SoC : 40% - 60%

Common Floorplan Issues & How To Solve

Issues (If Incorrect)	How To Solve
✗ High core utilization (>70%) → Congestion, timing degrade.	✓ Increase die size or reduce core utilization (40-70%).
✗ Low core utilization (<40%) → Wastage of silicon area.	✓ Reduce die size or increase utilization.
✗ Bad aspect ratio (<1 or >2) → Poor routability.	✓ Maintain AR between 1 to 2.
✗ Insufficient routing channels → Congestion, detours.	✓ Reserve proper H & V channels during FP.
✗ Macros too close / unbalanced → Heat / IR drop / timing.	✓ Maintain spacing & balance macro placement.
✗ Improper power ring / stripes → IR drop issues.	✓ Add/optimize power rings and stripes.
✗ IOs clustered on one side → Routing congestion.	✓ Distribute IOs on all sides uniformly.

Key Tips

- Plan early, verify early.
- Keep utilization balanced.
- Reserve routing resources.
- Plan power from the beginning.
- Check DRC after every major change.

Summary

- Floorplanning sets the stage for entire Physical Design.
- Good floorplan → better routing & timing closure.
- Follow design rules and tool guidelines.

Author : Konda Tharun Vivekananda

PD FLOOR PLANNING

5. MACRO AND MACRO PLACEMENT GUIDELINES

* WHAT IS MACRO ?

- Macro is a large pre-designed block that is placed as a single entity in the floorplan.
- Examples : SRAM, PLL, CPU, Analog, DSP, Memories, IP blocks, Hard macros.
- Macros are delivered with LEF abstracts (geometry + pins).
- They consume large area compared to standard cells and cannot be broken into cells.

* WHY WE NEED TO PLACE MACROS FIRST ?

- Macros are big and have fixed outlines.
- They consume most of the die area.
- They have strict placement rules and keep-out requirements.
- Early placement gives enough space for routing channels and standard cells.
- Helps in timing, power and congestion optimization.
- Avoids last minute re-floorplan which is costly.
- Ensures proper power, clock and signal connectivity.
- Improves routability and timing closure.

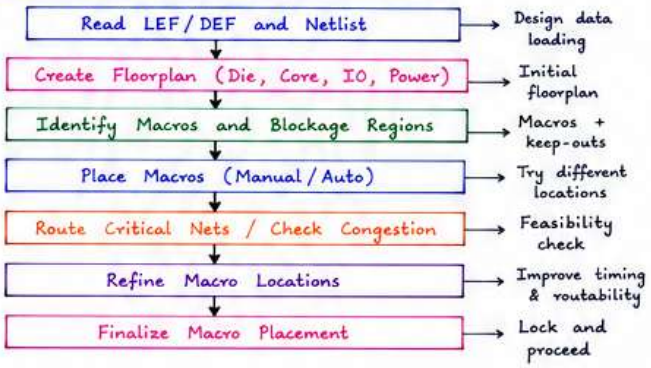
*** KEY POINT ***

Macro placement is one of the most critical steps in floorplanning and strongly affects PPA.

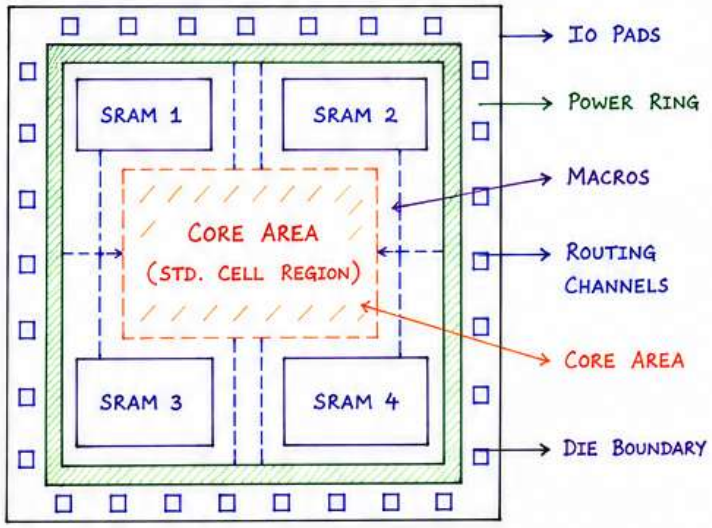
TYPES OF MACROS

TYPE	DESCRIPTION
HARD MACRO	Pre-designed and fixed layout. Delivered with LEF.
SOFT MACRO	Synthesizable block. Implemented during design.
BLOCK / BLACK BOX	Functional view only, no internal implementation.
IP MACRO	Third party IP like DDR, PCIe, USB, SerDes, PLL etc.

MACRO PLACEMENT FLOW (CONCEPTUAL)



MACRO PLACEMENT EXAMPLE (TOP VIEW)

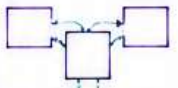


MACRO PLACEMENT GUIDELINES (8 RULES)

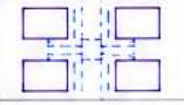
1. KEEP CENTER FREE
   → Keep the center area free for standard cells.



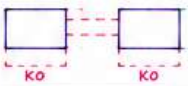
2. PLACE NEAR RELATED LOGIC
   → Place macros close to the logic they connect to most.



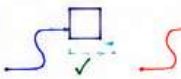
3. PROVIDE SUFFICIENT CHANNELS
   → Maintain enough horizontal and vertical routing channels.



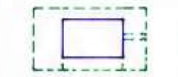
4. RESPECT KEEP-OUT & SPACING
   → Follow keep-out regions and maintain required spacing.



5. SHORT & DIRECT CONNECTIONS
   → Try to minimize wire length for critical connections.



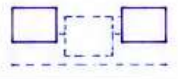
6. EASY POWER DISTRIBUTION
   → Place macros near power rings or power supplies.



7. AVOID CORNERS FOR BIG MACROS
   → Do not place large macros in corners; causes congestion.



8. LEAVE SPACE FOR FUTURE ECO
   → Keep extra space for buffer, ECO and routing resources.



PRACTICAL OBSERVATIONS

- Poor macro placement → congestion, long wires, timing fails.
- Too many macros in center → routing blockage.
- Ignoring spacing / keep-out → DRC violations later.
- Good macro placement → better timing, power & routability.
- Try multiple placements and compare QoR.

VIVA QUESTIONS

Q1. Why do we place macros before standard cells ?
→ Because macros are big, fixed and occupy more area. Early placement ensures enough space and routing.

Q2. What happens if macros are placed in the corners ?
→ It causes congestion, long interconnects and timing degradation.

*** KEY POINT ***

Macro placement is the backbone of good floorplan. It decides routability, timing, power and final QoR.

SUMMARY POINTS

- Macro is a large pre-designed block with fixed outline.
- We place macros first to reserve area and routing.
- Follow placement guidelines for better PPA.
- Keep center free and maintain sufficient channels.
- Good macro placement avoids costly floorplan changes.

AUTHOR : Konda Tharun Vivekananda

PD FLOOR PLANNING

6. KEEP-OUT MARGINS (HALO's)

* WHAT IS HALO / KEEP-OUT ?

- Halo (also called keep-out or blockage margin) is an empty space kept around a macro or block where no standard cells or routing are allowed.
- It creates a safe buffer zone for the macro.
- Defined during floorplanning.

* WHY DO WE USE HALO ?

- To avoid routing congestion near macros.
- To meet timing by giving routing detours.
- To provide power grid connectivity around macros.
- To ease IR drop and electromigration issues.
- To provide space for tap cells, decaps, buffers.
- To avoid DRC / LVS violations near macros.
- To allow ECO and rework without disturbing macros.
- To protect macros from heat and coupling.

* HOW HALO IS DEFINED ?

- Specified in all 4 sides : Top, Bottom, Left, Right.
- Can be symmetric or asymmetric.
- Typical values depend on technology and macro size.

```
set_halo { <object> } -top <t> -bottom <b>
   -left <l> -right <r> -halo_type hard
   -halo_margin {<t> <b> <l> <r>}
```

* ISSUES IF HALO IS INCORRECT

1. No or very small halo	→ High congestion → Long wires & detours → Timing degradation → IR drop failures
2. Unequal / poor halo	→ Unbalanced routing → IR drop locally high → EM violations
3. Too large halo	→ Waste of silicon area → Low utilization → Higher cost
4. Missing power access	→ Power grid blockage → IR drop, droop issues
5. No space for decaps/buffers	→ Local noise issues → Poor performance
6. No space for ECO	→ Difficult rework → Macro obstruction

* KEY POINT *

Halo is a small empty space around macros that protects timing, power, routability and reliability. Proper halo improves QoR and avoids costly rework.

VIVA QUESTIONS

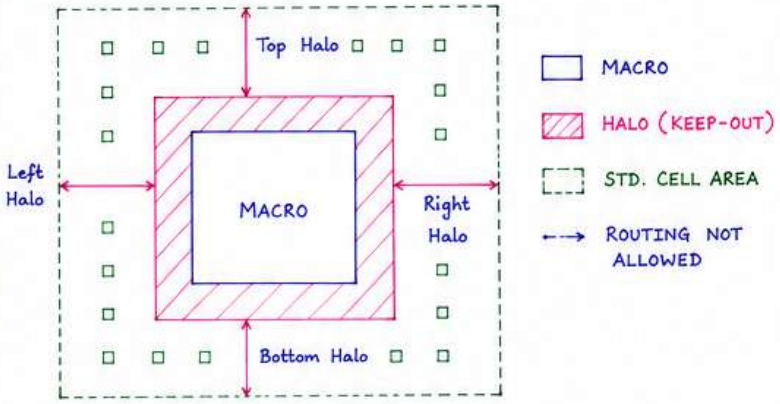
Q1. What is Halo in Physical Design ?

→ It is the keep-out margin around macros or blocks where no cells or routing are allowed.

Q2. Why do we use Halo around macros ?

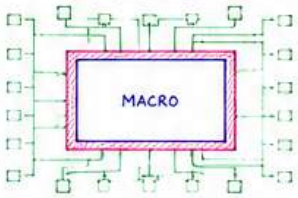
→ To reduce congestion, meet timing, ease IR drop, provide power, allow ECO and avoid DRC issues.

HALO ILLUSTRATION



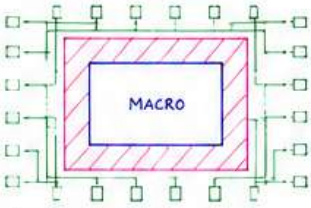
EFFECT OF HALO (EXAMPLES)

A) HALO TOO SMALL



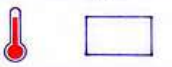
- x High congestion around macro
- x Long detours & timing impact
- x Routing may fail
- x IR drop & EM risk high

B) HALO OPTIMAL



- ✓ Good routing resources
- ✓ Better timing & power
- ✓ Lower congestion
- ✓ Higher reliability & QoR

MACRO PLACEMENT GUIDELINES (8 RULES)

1. Keep macros center free. → Standard cells need space in the center.	 ✓
2. Place macros near related logic. → Keep communication distance small.	 ✓
3. Align macros to rows / grid. → Easier routing & sign-off.	 ✓
4. Maintain sufficient halo on all sides. → Follow technology & design rules.	 ✓
5. Provide easy power connectivity. → Connect macros to rings / stripes.	 ✓
6. Avoid blocking routing channels. → Keep H & V channels open.	 ✓
7. Avoid corners for large macros. → Corners reduce routing capacity.	 ✓ X
8. Keep thermal / noisy blocks away. → Avoid heat & noise coupling.	 ✓

PRACTICAL OBSERVATIONS

- * Start with coarse macro placement → refine later.
- * Always obey technology min halo requirements.
- * Use congestion maps to adjust halo and macro locations.
- * Balanced macro placement reduces timing & power issues.

TYPICAL HALO VALUES (EXAMPLE)

Small Macro	: 2 - 5 μm
Medium Macro	: 5 - 10 μm
Large Macro	: 10 - 20 μm

(Depends on tech & design)

SUMMARY POINTS

- Halo is a keep-out margin around macros.
- It improves timing, power, routability and reliability.
- Incorrect halo → congestion, timing fails, IR drop, routing issues.
- Follow 8 macro placement guidelines for good floorplan.
- Proper halo and macro placement → better QoR.

AUTHOR : Konda Tharun Vivekananda

PD FLOOR PLANNING

FLYWIRE ANALYSIS

✱ WHAT IS FLYWIRE ANALYSIS ?

- Flywire (also called airwire) is a temporary straight line connection used to represent nets in the design before actual routing.
- Flywire analysis estimates the routability of the design after floorplan / macro placement.
- It helps to identify long nets, congestion hot-spots and routing difficult areas early.

WHY DO WE USE FLYWIRE ANALYSIS ?

- To check if the design is routable or not.
- To identify congestion and critical long nets.
- To guide macro placement / re-arrangement.
- To reduce ECOs and timing violations later.
- To estimate wirelength, timing and IR drop early.

ISSUES IF INCORRECT / POOR FLYWIRE QUALITY

- ✗ Many long flywires → High wirelength & RC delay
- ✗ High congestion → Routing detours & failures
- ✗ Timing violations → Setup / hold failures
- ✗ IR drop & EM problems due to long wires
- ✗ More ECOs and re-work → Cost & time increase
- ✗ May result in DRC / antenna violations later

✱ KEY POINT ✱

Good flywire quality = shorter length, balanced distribution and less congestion → better routability, timing, power and QoR.

TYPES OF FLYWIRES

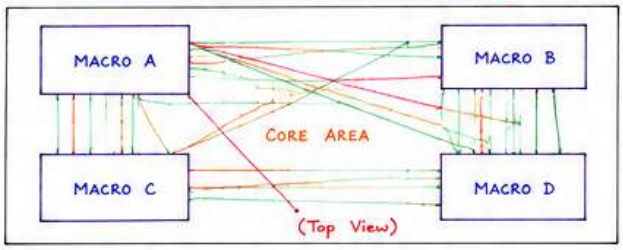
- Global Flywire : Across the entire design.
- Block / Macro Flywire : Within a macro or between related blocks.
- Top-level Flywire : Between major blocks / IOs.

HOW FLYWIRE IS GENERATED ?

- After floorplan and macro placement.
- Tool connects all logical nets using straight lines (ignoring obstacles).
- It does not follow routing rules (layers/vias).
- Used only for estimation and analysis.

FANOUT BASED FLYWIRE VIEW

Low Fanout Medium Fanout High Fanout



Green → Low fanout nets (few connections)
Orange → Medium fanout nets (moderate connections)
Red → High fanout nets (many connections)

VIVA QUESTIONS

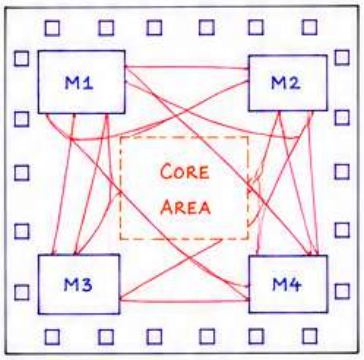
Q1. What is flywire analysis ?
→ It is a pre-routing estimate using straight line connections to check routability, congestion and timing early

Q2. Why flywire length is important ?
→ Long flywires cause high delay, congestion, IR drop and routing failures.

Q3. What action is taken if flywires are poor ?
→ Optimize macro placement, fix congestion and re-run analysis until routability is good.

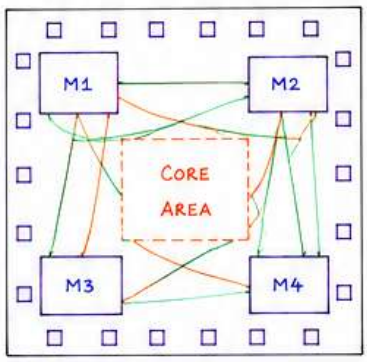
EXAMPLE : FLYWIRE BEFORE & AFTER IMPROVEMENT

A) POOR FLYWIRE (BEFORE)



- ✗ Very long flywires
- ✗ Crossing over the core
- ✗ High congestion expected
- ✗ Poor routability
- ✗ Higher delay & IR drop

B) GOOD FLYWIRE (AFTER)



- ✓ Shorter flywires
- ✓ Minimal crossings
- ✓ Low congestion
- ✓ Good routability
- ✓ Better timing & power

FANOUT vs FLYWIRE LENGTH

FANOUT	DESCRIPTION	IMPACT ON FLYWIRE	ACTION
LOW (1 - 3)	Net connects few pins.	Short flywire, easy to route.	No major issue.
MEDIUM (4 - 10)	Net connects to moderate pins.	Moderate length, check congestion.	Ensure balance distribution.
HIGH (> 10)	Net connects to many pins.	Very long flywire, causes congestion and timing issues.	Use buffers, re-structure, re-place macros.

TOOL COMMANDS (EXAMPLE)

- create_flywire -selection all_nets → Create flywires
- report_flywire -summary → Summary report
- report_flywire -max_length <value> → Find long nets
- report_congestion → Check congestion
- remove_flywire → Clear flywires
- analyze_flywire → Analyze quality

PRACTICAL OBSERVATIONS

- ✱ If most flywires are red (long) → routing will be very difficult.
- ✱ Check congestion early and move macros / blocks.
- ✱ Use power rings/stripes before final placement for better results.
- ✱ Keep critical macros close to their logic to reduce wirelength.
- ✱ Flywire analysis should be done iteratively after each change.

FLYWIRE QUALITY CHECKLIST

- ✓ Total flywire length is minimized
- ✓ Average flywire length is within target
- ✓ Maximum flywire length is within limit
- ✓ No excessive crossings in core area
- ✓ Congestion is low and evenly distributed
- ✓ Critical nets have short flywires
- ✓ Macros are logically placed
- ✓ Power / clock network connectivity is healthy

SUMMARY POINTS

- Flywire is a temporary straight line connection before routing.
- It helps to check routability, congestion and timing early.
- Good flywire = short length, less crossing, low congestion.
- Always fix poor flywires before proceeding to placement.
- Better flywire quality leads to better QoR and less ECO.

KEY BENEFITS OF GOOD FLYWIRE ANALYSIS

- ✱ Improves routability
- ✱ Reduces congestion
- ✱ Better timing estimation
- ✱ Lowers IR drop & EM risk
- ✱ Reduces ECOs & re-spins
- ✱ Saves time and cost
- ✱ Helps in macro placement
- ✱ Better power & signal integrity
- ✱ Improves final design QoR

AUTHOR : Konda Tharun Vivekananda

PD FLOOR PLANNING

6. IO AND PIN PLANNING

✳ WHAT IS IO AND PIN PLANNING ?

- IO (Input/Output) planning decides the location, type and distribution of IO pads on the die boundary.
- Pin planning decides exact pin locations and connectivity to internal blocks / macros.
- Done during floorplanning before placement.

✳ WHY DO WE NEED TO PLAN IO AND PINS ?

- To provide communication between chip and outside.
- To minimize wirelength and routing congestion.
- To balance signal delay, timing and power.
- To meet design rule checks (spacing, density, access).
- Proper pin planning improves routability and QoR.

✳ ISSUES IF IO / PIN PLANNING IS INCORRECT

✗ Long interconnects	→	High delay & wirelength
✗ Routing congestion	→	Detours & routing failures
✗ Timing violations	→	Setup / Hold fails
✗ IR drop at IO bank	→	Power supply problems
✗ Unbalanced IO sides	→	Poor utilization
✗ Crosstalk / SI issues	→	Signal integrity loss
✗ ECO becomes costly	→	Floorplan changes required
✗ Package / board issues	→	Difficult bonding / escape

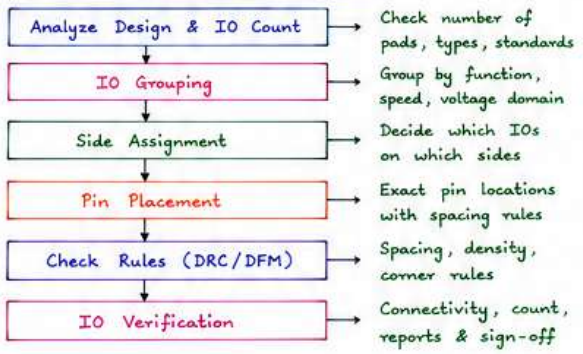
✳ KEY POINT ✳

Good IO and pin planning early in floorplanning reduces wirelength, congestion, timing risk and ECO cost.

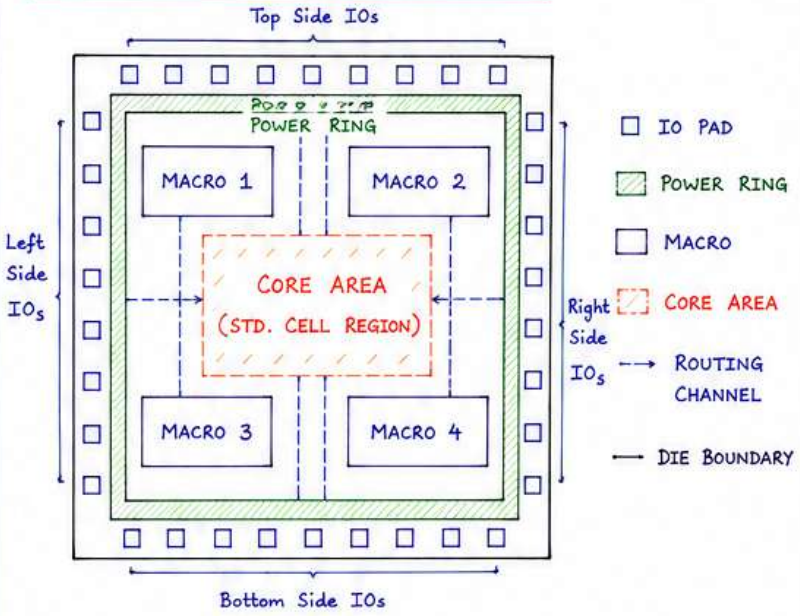
TYPES OF IOS

- Digital IO : General purpose digital signals.
- Analog IO : Sensitive analog signals.
- Power / Ground IO : VDD, VSS, AVDD, AVSS.
- Special IO : Test, JTAG, scan, clock, reset.

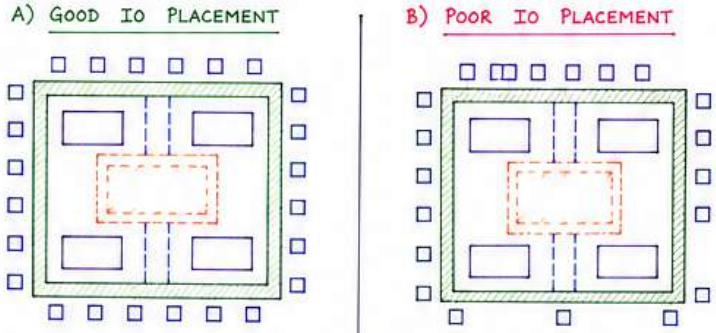
IO PLACEMENT FLOW (CONCEPTUAL)



IO DISTRIBUTION EXAMPLE (TOP VIEW)



GOOD VS POOR IO PLACEMENT



✓ Good IO Placement	✗ Poor IO Placement
✓ Balanced IOs on all sides	✗ Unbalanced IOs
✓ Short interconnects	✗ Long wires to other sides
✓ Uniform routing channels	✗ Congestion on few sides
✓ Better routability & timing	✗ Timing & power issues

IMPORTANT GUIDELINES FOR IO AND PIN PLANNING

1. Distribute IOs uniformly on all four sides.
2. Place high speed IOs close to the package edge.
3. Keep analog IOs separate from noisy digital IOs.
4. Place power and ground pads at regular intervals.
5. Maintain minimum spacing between IOs as per DRC.
6. Reserve space for future ECO / additional pins.
7. Consider routing channel availability while placing IOs.
8. Follow technology / package pin mapping rules.

PRACTICAL OBSERVATIONS

- ✳ Good IO planning saves lot of effort in routing stage.
- ✳ Early placement of critical IOs helps in better timing.
- ✳ Always verify pin accessibility and channel width.
- ✳ Use tool reports to check IO density and imbalance.

VIVA QUESTIONS

Q1. Why is IO planning done before placement ?
→ Because it decides external connectivity and affects routing channels, block placement and timing.

Q2. What happens if IOs are not balanced on sides ?
→ It causes long wires, poor utilization and routing congestion on few sides.

Q3. Why analog IOs are placed separately ?
→ To avoid digital noise and crosstalk and to meet signal integrity.

IO / PIN CHECKLIST

- ☑ IO count matches spec
- ☑ Correct side assignment
- ☑ Min spacing rule met
- ☑ Power / ground pins properly distributed
- ☑ High speed IOs placed close to package edge
- ☑ DRC clean
- ☑ Reports reviewed

BENEFITS OF GOOD IO / PIN PLANNING

- ✓ Reduces wirelength
- ✓ Less congestion
- ✓ Better timing closure
- ✓ Lower power
- ✓ Fewer ECOs
- ✓ Improved overall QoR

SUMMARY POINTS

- IO & pin planning defines chip boundary communication.
- Done during floorplanning before placement.
- Good planning ensures routability, timing and power.
- Balanced and rule-compliant IO placement avoids costly changes.

AUTHOR : Konda Tharun Vivekananda

PD FLOOR PLANNING

7. POWER PLANNING

* INTRODUCTION

- Power planning is the process of designing low resistance power distribution network (PDN) to deliver stable VDD & GND to all parts of the chip.
- It includes power rings, power stripes (rails) and power vias.
- Goal → Low IR drop, low EM, reliable & uniform power delivery.

* WHY DO WE DO POWER PLANNING ?

- To provide low impedance path for power.
- To reduce IR drop across the chip.
- To prevent electromigration (EM) failure.
- To ensure uniform power to std cells, macros & IOs.
- To meet timing by avoiding voltage droop.
- To improve reliability and signal integrity.

ISSUES IF INCORRECT / POOR POWER PLANNING

x	High IR drop	→ Voltage droop, timing failure
x	EM violations	→ Metal degradation / opens
x	Power grid hotspots	→ Local over current
x	Uneven voltage	→ Functional failures
x	Excessive noise	→ Signal integrity problems
x	Routing blockages	→ Affects routability
x	DRC / spacing issues	→ Sign-off failures

HOW TO AVOID / SOLVE THESE ISSUES

- ✓ Use wide & thick power rings.
- ✓ Use sufficient number of power stripes with proper pitch.
- ✓ Follow foundry recommended width & spacing.
- ✓ Add enough vias (via arrays) between metals.
- ✓ Use multiple metal layers for power.
- ✓ Check IR drop & EM early and iterate.
- ✓ Balance power grid with timing & routability.

WHY POWER PLANNING IS DONE IN FLOORPLAN ITSELF ?

- → Because power network is a physical structure.
- → It consumes area and routing resources.
- → Influences macro placement, std cell placement and routing.
- → Early power planning avoids rework and congestion later.
- → It decides power pin locations on macros / IOs.
- → Helps accurate estimation of IR drop, EM and power.

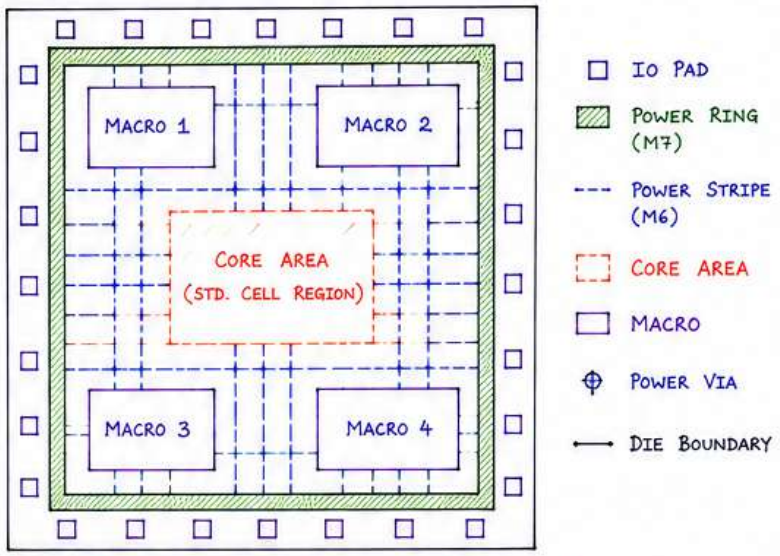
*** KEY POINT ***

Good power planning in floorplan stage ensures stable power, avoids IR drop & EM issues and leads to better timing, reliability and QoR.

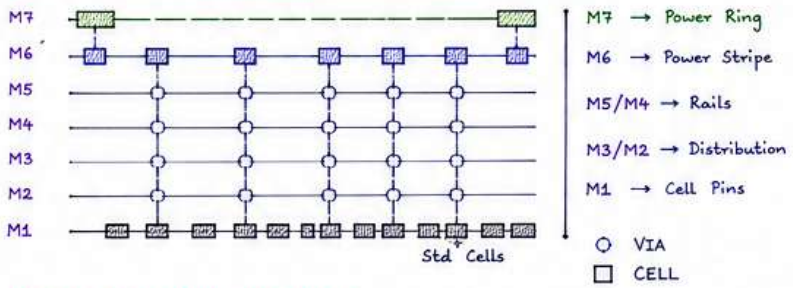
COMMON POWER NETWORK COMPONENTS

COMPONENT	DESCRIPTION
POWER RING	Closed ring around core using Mx layer.
POWER STRIPE (POWER RAIL)	Vertical / Horizontal metal lines inside core connecting rings.
POWER VIA	Vias connecting different metal layers.
POWER PAD / PIN	Connection from die to package / PCB.
DECOUPLING CAP	Reduces noise and supplies transient current.

POWER PLANNING EXAMPLE (TOP VIEW)



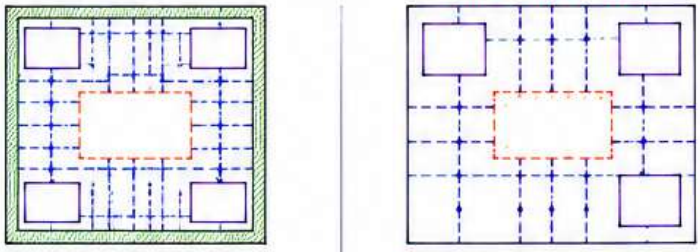
POWER RING & STRIPE STRUCTURE (SIDE VIEW)



POWER PLANNING CHECKLIST

- ☑ Use sufficient power ring width
- ☑ Use enough number of stripes
- ☑ Maintain recommended pitch
- ☑ Ensure power ring continuity
- ☑ Add enough vias between layers
- ☑ Connect macros to ring / stripes
- ☑ Check IR drop early
- ☑ Check EM violations
- ☑ Follow foundry DRC rules
- ☑ Balance power with routing

GOOD vs POOR POWER PLANNING



- ✓ Uniform power distribution
- ✓ Low IR drop
- ✓ Low EM risk
- ✓ Better timing & reliability

- x Uneven power distribution
- x High IR drop
- x High EM risk
- x Timing failures & rework.

PRACTICAL OBSERVATIONS

- * Start with coarse power planning → refine during detailed stages.
- * Use tool reports (IR drop, EM) to verify and fix early.
- * More stripes & wider metals → better power but more area.
- * Keep power pins close to macros to reduce voltage drop.

VIVA QUESTIONS

Q1. Why is power planning important ?
→ It ensures stable power, reduces IR drop, meets timing and improves reliability.

Q2. What happens if IR drop is high ?
→ Voltage droop, timing failures, functional failures.

Q3. Why power planning is done in floorplan itself ?
→ Because it affects area, placement, routing and avoids costly rework later.

BENEFITS

- Low IR drop
- Low EM risk
- Better timing closure
- Higher reliability
- Uniform voltage
- Better QoR

SUMMARY POINTS

- Power planning ensures reliable power delivery across the chip.
- It includes power rings, power stripes and vias eady.
- Incorrect power planning causes IR drop, EM and timing failures.
- It is done in floorplan stage to avoid costly rework.
- Follow foundry rules and verify using IR drop & EM reports.

AUTHOR : Konda Tharun Vivekananda

PD FLOOR PLANNING

8. Special Cells During Floorplan

* Introduction

- Special cells are non functional (or helper) cells used in the design to meet physical, reliability and sign-off requirements.
- These cells are placed during floorplan stage itself to reserve space, ensure connectivity and avoid problems later.

* Why Do We Use Special Cells in Floorplan ?

- To reserve space and avoid congestion in later stages.
- To provide power integrity and ESD protection.
- To meet DRC / LVS / Antenna / EM rules.
- To improve reliability and manufacturability.
- Early placement avoids costly ECOs and re-routes.

* Issues If Special Cells Are Not Placed Early / Incorrectly

- ✗ Congestion in placement & routing → Detours
- ✗ IR Drop / Voltage Drop → Timing failures
- ✗ ESD damage risk → Reliability issues
- ✗ Antenna violations → Metal shorts / opens
- ✗ DRC / LVS violations → Sign-off failure
- ✗ Late ECOs → Area, time & cost increase

* Why Done In Floorplan Itself ?

- Floorplan defines die area, macros and blockages.
- Special cells require reserved locations and connections (power / ground).
- Early planning ensures uniform distribution and meets design rules.
- Prevents routing congestion and rework later.
- Helps accurate estimation of PPA and QoR.

*** Key Point ***

Place special cells during floorplan to reserve space, ensure connectivity, meet design rules and avoid late-stage rework & ECOs.

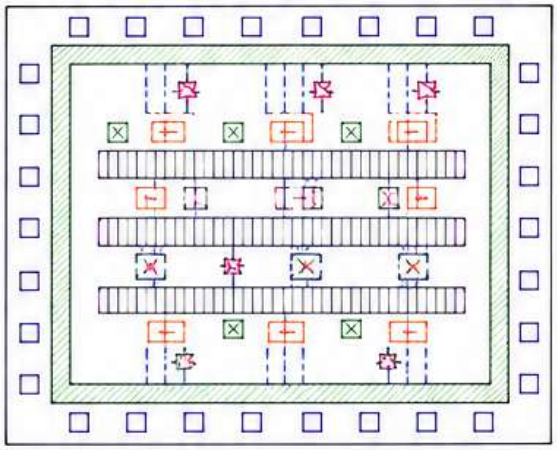
Common Special Cell Categories

- ESD protection cells
- Decoupling / Bypass capacitors
- Well tap / Substrate tap cells
- Antenna diode cells
- Tie cells (Tie-high / Tie-low)
- End cap cells
- Spare cells
- Filler cells

Types of Special Cells Used in Floorplan

S.No.	Special Cell	Why We Use	Situation / Purpose	Diagram (Top View)
1.	ESD Protection Cell	Protects I/O and internal circuits from ESD discharge.	Placed near I/O pads and sensitive blocks.	 ESD Cell
2.	Decoupling Capacitor (Decap)	Reduces IR drop by supplying instant current.	Distributed across core and around high switching blocks.	 Decap
3.	Well Tap / Substrate Tap Cell	Provides substrate bias and prevents latch-up.	Placed at regular intervals in large core area.	 Tap Cell
4.	Antenna Diode Cell	Prevents antenna effect during metal etch.	On long nets with high metal area to gate area.	 Antenna Diode
5.	Tie High / Tie Low Cell	Fixes unused inputs to logic '1' or '0'.	Used for default connections and unused inputs.	 Tie-Hi, Tie-Lo
6.	End Cap Cell	Provides proper well / implant termination.	At the end of standard cell rows.	 End Cap
7.	Spare Cell	Allows post-silicon ECO / repair.	Distributed in rows for future use.	 Spare Cell
8.	Filler Cell	Fills small gaps in rows to meet density.	In small gaps during placement.	 Filler Cell

Floorplan View With Special Cells



How To Avoid Issues ?

- ✓ Follow foundry rules and spacing.
- ✓ Use recommended pitch & intervals.
- ✓ Distribute cells uniformly.
- ✓ Maintain power ring / stripe continuity.
- ✓ Check IR drop, EM and antenna early.
- ✓ Verify DRC and LVS continuously.

Practical Observations

- * Always start with power ring first.
- * Reserve space for special cells early.
- * Do not ignore tap & decap distribution.
- * More decaps near high switching areas.
- * Tie cells reduce floating input issues.
- * Spare cells save chip in post-silicon.

Viva Questions

Q1. Why are special cells placed in floorplan stage ?
→ To reserve space, ensure connectivity and meet physical, reliability and sign-off requirements.

Q2. What happens if special cells are not placed early ?
→ Congestion, IR drop, DRC/LVS failures and costly ECOs occur.

Q3. Give some examples of special cells.
→ ESD cells, decap cells, well tap, antenna diodes, tie cells, end caps, spare cells, filler cells.

Q4. Why do we need filler cells ?
→ To fill small gaps for uniform rows, improve density and reduce DRC violations.

Benefits of Using Special Cells Early

- ✓ Stable power & low IR drop
- ✓ Better reliability & ESD safety
- ✓ Meets DRC / LVS / Antenna rules
- ✓ Less congestion & better routability
- ✓ Avoids late ECOs & cost
- ✓ Improved timing & overall QoR

Summary Points

- Special cells are helper cells for physical, reliability and sign-off.
- Place them during floorplan to reserve space and ensure uniform distribution.
- Helps to avoid congestion, IR drop, DRC, antenna and EM issues.
- Follow foundry rules and tool reports for correct planning.
- Early planning gives better QoR and reduces rework.

PD FLOOR PLANNING

9. CONGESTION PLANNING IN FLOORPLAN

✱ INTRODUCTION

- Congestion planning is the process of reserving enough routing resources in floorplan stage based on expected (upcoming) routing demand.
- By providing proper routing channels and spacing, we avoid congestion, detours and timing degradation later.
- It is easier and cheaper to fix congestion in floorplan than in placement / routing stages.

✱ WHY DO WE DO CONGESTION PLANNING ?

- To reserve space for high routing demand nets.
- To ensure smooth routing without detours.
- To meet timing by reducing extra wirelength & delay.
- To avoid routing congestion and routing failures.
- To minimize DRC / antenna and via stack issues.
- To improve overall QoR and reduce ECOs.

✱ ISSUES IF NOT DONE WELL

✗ Routing congestion	→	Detours, longer wires
✗ Timing violations	→	Setup / Hold failures
✗ High IR drop	→	Voltage drop & failures
✗ More vias / via stacks	→	DRC / reliability risk
✗ Antenna violations	→	Added diodes / area cost
✗ Routing capacity exceeded	→	Routing failure
✗ ECO cost increases	→	More iterations, delay
✗ Power grid blockage	→	Hotspots, reliability issues

✱ HOW TO AVOID / SOLVE CONGESTION ISSUES

- ✓ Analyze routing demand early (global routing / routes).
- ✓ Provide sufficient routing channels & blockages spacing.
- ✓ Widen channels for high demand nets (clk, buses, etc.).
- ✓ Follow design rules for macro spacing and blockages.
- ✓ Balance placement regions and keep critical nets short.
- ✓ Iterate congestion heatmaps and adjust floorplan.

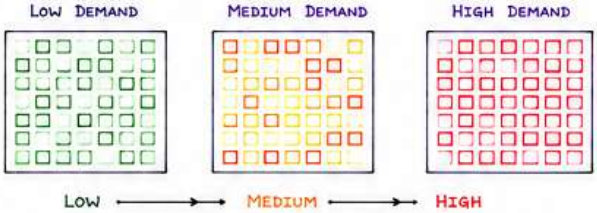
WHY CONGESTION PLANNING IS DONE IN FLOORPLAN ?

- → Floorplan stage is before placement & routing.
- → At this stage, we can allocate space, channels and blockages with minimum cost.
- → After placement, moving macros or blockages is costly.
- → Good floorplan → good placement → easy routing → better timing, power and QoR.

✱ KEY POINT ✱

Provide routing resources early in floorplan to match upcoming routing demand. Good congestion planning avoids detours, timing failures and ECO cost.

ROUTING DEMAND LEVELS (CONCEPTUAL)



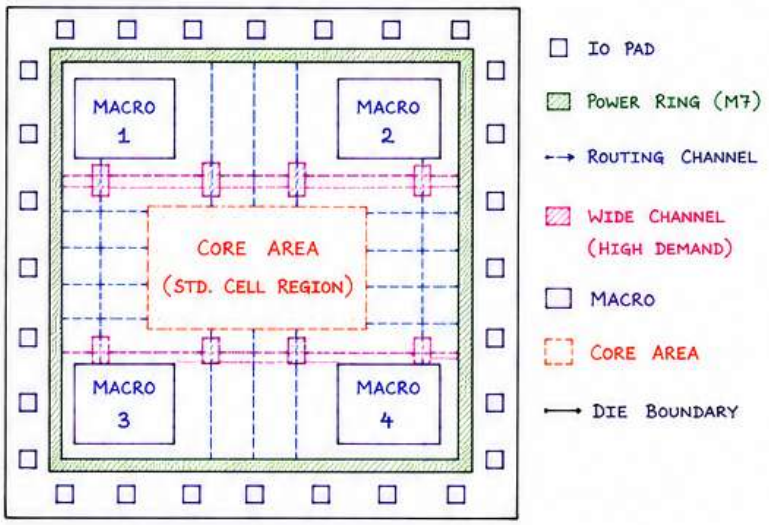
VIVA | QUESTIONS

Q1. Why is congestion planning done in floorplan stage ?
→ Because it is easier to allocate space and routing resources early before placement and routing.

Q2. What happens if congestion planning is not good ?
→ Congestion, long detours, timing failures, high IR drop, routing failures and more ECO cost.

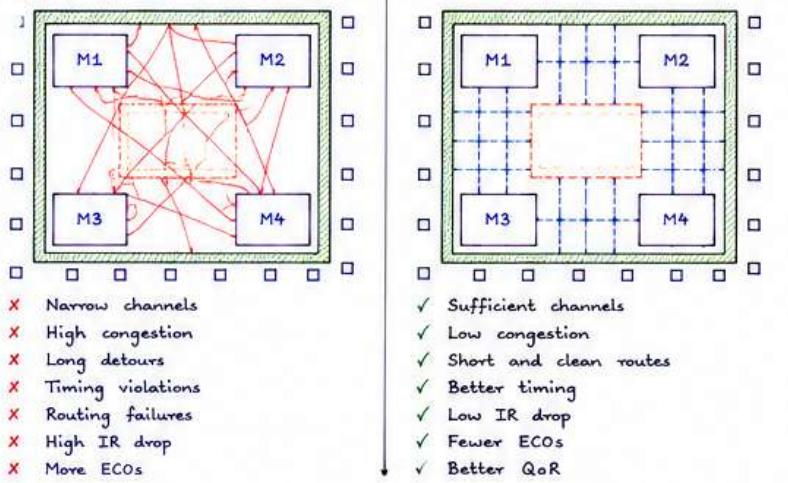
Q3. Which nets need more routing resources ?
→ Clock nets, wide buses, high fan-out control nets, memory interfaces, global signals.

CONGESTION PLANNING EXAMPLE



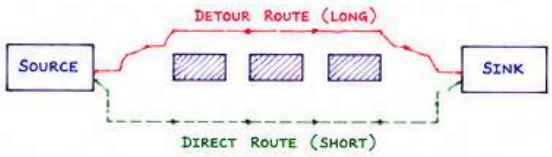
Reserve wider channels for high demand nets (clk, buses, etc.)

IF CONGESTION PLANNING IS INCORRECT



A) Poor Planning (Less Space)	B) Good Planning (More Space)
✗ Narrow channels	✓ Sufficient channels
✗ High congestion	✓ Low congestion
✗ Long detours	✓ Short and clean routes
✗ Timing violations	✓ Better timing
✗ Routing failures	✓ Low IR drop
✗ High IR drop	✓ Fewer ECOs
✗ More ECOs	✓ Better QoR

IF INCORRECT ROUTING DUE TO CONGESTION



EFFECTS

- ✗ More wirelength
- ✗ More delay
- ✗ Timing failures
- ✗ More vias
- ✗ Higher power
- ✗ Lower reliability

CONGESTION ANALYSIS TOOLS (USED EARLY)

- Global Routing (GR) ..... Estimated routing demand
- Congestion Heatmap ..... Identify hot spots
- Route Overflow Report ..... % overflow in regions
- Capacity vs Demand Check ... Ensure routing capacity



PRACTICAL OBSERVATIONS

- ✱ Start with global routing and identify hot spots.
- ✱ Reserve more space for clock, buses and memory nets.
- ✱ Keep macro spacing adequate to allow multiple metal layers.
- ✱ Do not ignore routing resource planning in floorplan.
- ✱ A little extra space in floorplan saves huge cost later.

BENEFITS

- ✓ Low congestion
- ✓ Better timing closure
- ✓ Low IR drop & EM
- ✓ Fewer DRC / Antenna
- ✓ Smooth routing
- ✓ Better overall QoR

SUMMARY POINTS

- Congestion planning reserves routing resources based on upcoming demand.
- Poor planning → congestion, detours, timing violations and more ECOs.
- Done in floorplan stage to reduce cost and ensure QoR.
- Use GR, heatmaps and reports to guide floorplan decisions.
- Good congestion planning leads to easy routing and reliable design.

AUTHOR : Konda Tharun Vivekananda

PD FLOOR PLANNING

10. BLOCKAGES IN FLOORPLAN

* INTRODUCTION

- Blockages are reserved areas in the floorplan where tr placement or routing is restricted or prohibited.
- They help to protect structures, fix routing channels and guide tools for better QoR.

* WHY DO WE USE BLOCKAGES ?

- To protect macros, IOs, power rings and fixed structures.
- To reserve space for routing channels and buses.
- To avoid congestion and routing detours.
- To follow DRC, LVS, antenna and design rules.
- To fix ECO / DFT / Clock routing regions in advance.

* ISSUES IF BLOCKAGES ARE NOT USED / INCORRECT

x	Cells placed over sensitive areas	→ DRC / LVS fail
x	No routing channels	→ Congestion, detours
x	Antenna / EM / IR issues	→ Reliability problems
x	Late ECO needs more area	→ Cost & schedule slip
x	Timing violation due to long routes	→ Setup / Hold fail
x	Metal density / slot violations	→ Sign-off failures
x	Power ring / via damage	→ IR drop, EM risk

* HOW TO USE BLOCKAGES EFFECTIVELY ?

- ✓ Identify all fixed structures and sensitive areas early.
- ✓ Reserve adequate space for routing (M6/M7 channels).
- ✓ Maintain proper spacing from macros / power rings.
- ✓ Use correct blockage type for correct purpose.
- ✓ Review with DRC, congestion map and design rules.
- ✓ Update blockages during ECO and re-verify.

*** KEY POINT ***

Correct blockage planning in floorplan saves routing resources, avoids congestion and helps to achieve better timing, reliability and sign-off.

TYPES OF BLOCKAGES

S.NO.	TYPE	DESCRIPTION	ALLOWED / DISALLOWED	USE CASE
1.	HARD BLOCKAGE	No cell placement and no routing allowed.	Disallow Cells Routing	Macros, IOs, Power rings, Analog blocks, SRAM, PLL
2.	SOFT BLOCKAGE	No cell placement but routing is allowed.	Disallow Cells Allow Routing	Keep-out for placement, but allow signal routing.
3.	PARTIAL BLOCKAGE (DENSITY)	Cells are allowed but with limited density / capacity.	Allow Cells (with limit) Allow Routing	Reserve space for routing, clock tree, local congestion control.
4.	ROUTING BLOCKAGE	No routing allowed but cell placement allowed.	Allow Cells Disallow Routing	Protect analog nets, noise sensitive nets, certain areas.

BLOCKAGE SELECTION GUIDE

- If structure is fixed and cannot be touched → Hard blockage
- If only cells should be kept away → Soft blockage
- If some cells allowed with limited density → Partial blockage
- If routing should be restricted only → Routing blockage

VIVA QUESTIONS

Q1. What is a blockage in floorplanning ?
→ Reserved area used to restrict or control placement and/or routing for better design implementation.

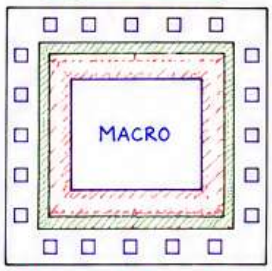
Q2. Why are hard blockages used ?
→ To protect fixed structures (macros, IOs, power rings) where neither cells nor routing is allowed.

Q3. Difference between soft blockage and hard blockage ?
→ Hard : No cells & no routing. Soft : No cells but routing is allowed.

Q4. Why do we use partial blockages ?
→ To limit cell density in a region and reserve space for routing or clock tree etc.

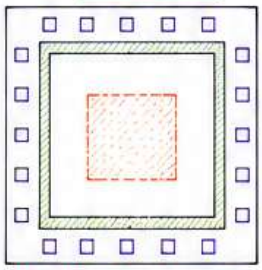
BLOCKAGE EXAMPLES WITH DIAGRAMS

1) HARD BLOCKAGE

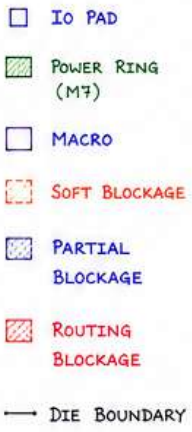


No cells & No routing allowed inside.

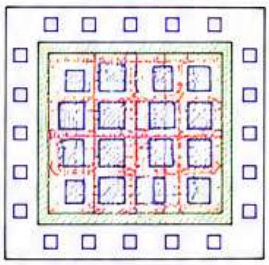
2) SOFT BLOCKAGE



No cells allowed inside, routing is allowed.

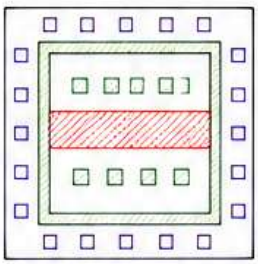


3) PARTIAL BLOCKAGE (DENSITY)



Cells allowed but with limited density (e.g. 30%). Routing allowed.

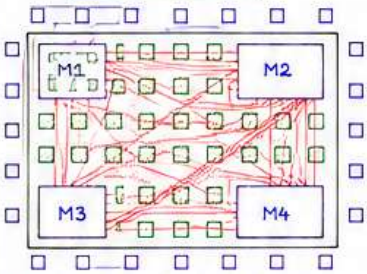
4) ROUTING BLOCKAGE



Cells allowed. Routing not allowed in blocked region.

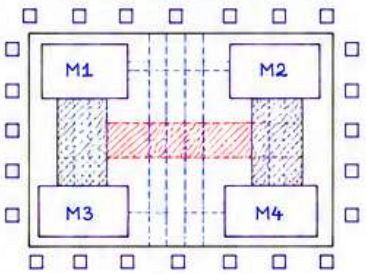
EFFECTS OF INCORRECT / MISSING BLOCKAGES

A) WITHOUT BLOCKAGES



- x Cells overlap fixed structures
- x Routing congestion & Detours
- x Long wires → High delay
- x DRC / Antenna violations
- x IR drop / EM problems
- x Bad timing & QoR

B) WITH CORRECT BLOCKAGES



- ✓ Fixed structures protected
- ✓ Clear routing channels
- ✓ Low congestion
- ✓ Shorter routes & less delay
- ✓ Fewer violations
- ✓ Better timing & QoR

COMMON BLOCKAGE PLACEMENT LOCATIONS

- Around macros / SRAM / analog blocks → use hard blockage.
- Around power rings → keep soft blockage for placement.
- Reserve M6/M7 channels between macros → partial blockage.
- Protect critical nets or analog regions → routing blockage.
- Reserve space for clock tree, tie cells, decoupling caps → partial.

CHECKLIST

- ✓ Identify all fixed blocks
- ✓ Define blockage type
- ✓ Apply correct spacing
- ✓ Review placement
- ✓ Check congestion
- ✓ Verify DRC / LVS
- ✓ Re-check after ECO

BENEFITS OF USING BLOCKAGES

* Protects important structures	* Fewer DRC / LVS violations
* Reduces routing congestion	* Saves ECO time and cost
* Improves timing & reliability	* Better overall QoR

SUMMARY POINTS

- Blockages guide tools to place cells and route wires correctly.
- Use correct type for correct purpose during floorplanning.
- Good blockage planning results in smooth routing and better sign-off.

PRACTICAL OBSERVATIONS

- * Always start with hard blockages for fixed structures.
- * Keep routing channels wide enough for future metal layers.
- * Use partial blockages to control density in critical regions.
- * Continuously check congestion heatmap and adjust blockages.
- * Update blockages whenever design changes or ECO happens.

AUTHOR : Konda Tharun Vivekananda

PD FLOOR PLANNING

11. FLOORPLANNING OPTIMIZATION

* INTRODUCTION

- Floorplanning optimization is the iterative process of refining the floorplan to achieve best PPA (Power, Performance, Area) and routability.
- It balances macro placement, power distribution, routing resources, timing and congestion.
- Done during floorplan stage itself before placement.

* WHY DO WE OPTIMIZE FLOORPLAN ?

- To minimize congestion and routing detours.
- To reduce IR drop and improve power integrity.
- To meet timing (shorter critical paths, less skew impact).
- To improve area utilization and reduce dead space.
- To avoid expensive ECOs and re-spins.
- To ensure DRC / antenna / LVS clean physical design.

* WHAT IF NOT DONE WELL ?

X	High congestion	→ Detours, long wires, timing fail.
X	IR drop / EM issues	→ Voltage drop, reliability risk.
X	Timing violations	→ Setup / Hold failures.
X	Poor QoR	→ More power, more area, low perf.
X	Late ECOs	→ Cost, schedule slip, rework.
X	Routing failures	→ Incomplete nets, DRC fail.

* WHAT IF DONE INCORRECTLY (EXAMPLE)

- Suppose macros are placed very close with narrow channels.
- Power ring on only one layer with no stripes.
- IOs placed randomly without considering routes.

Result :

→ Congested channels
→ Long detour routes
→ High IR drop near macros
→ More vias / layer change
→ Timing degradation & higher power.

* KEY POINT *

Well optimized floorplan → smooth routing → good timing, low power, less area and high QoR.

FLOORPLANNING OPTIMIZATION METRICS

- Congestion (global routing demand vs capacity)
- HPWL (Half Perimeter Wire Length)
- IR Drop (Voltage drop)
- Timing Slack (estimated)
- Power Grid Integrity
- Ut 'zation %
- DRC / Antenna clean.

VIVA QUESTIONS

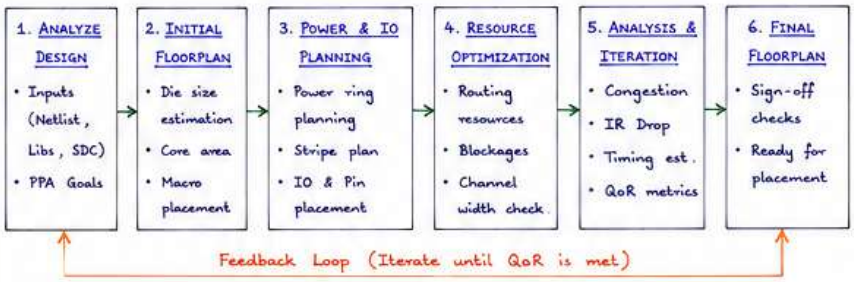
Q1. Why is floorplan optimization important ?
→ It ensures better routability, timing closure, power integrity and overall QoR with minimum cost & iterations.

Q2. What happens if macros are placed randomly ?
→ Congestion, long routes, high power and timing failures.

Q3. When do we start optimization in floorplan ?
→ During floorplan stage itself before placement.

Q4. What tools help in floorplan optimization ?
→ Global routing, congestion heatmaps, IR drop analysis, timing estimation and QoR reports.

FLOORPLANNING OPTIMIZATION FLOW (STAGE BY STAGE)

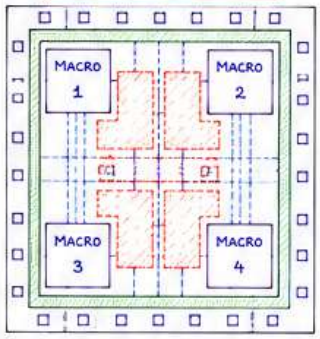


DETAILED STAGE EXPLANATION

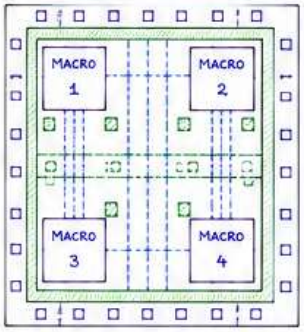
1) Analyze Design : Understand design size, macros, I/O count, power, clock, timing criticality and routing demand.
2) Initial Floorplan : Decide die size, core area, aspect ratio and place macros roughly to meet high level goals.
3) Power & IO Planning : Create power ring, stripes and vias plan. Place IOs & pins to reduce routing congestion.
4) Resource Optimization : Define routing channels, blockages, keep-outs and reserve space for special cells & clock tree.
5) Analysis & Iteration : Run global routing, congestion map, IR drop check, timing estimation and QoR reports. Iterate floorplan.
6) Final Floorplan : Freeze floorplan after meeting all goals and passes sign-off checks. Export DEF for next stage (Placement).

EXAMPLE – POOR vs OPTIMIZED FLOORPLAN

A) POOR FLOORPLAN



B) OPTIMIZED FLOORPLAN



IO PAD
POWER RING (M7)
POWER STRIPE (M4)
MACRO
ROUTING CHANNEL
CONGESTED AREA
BLOCKAGE
DIE BOUNDARY

A) Poor Floorplan	B) Optimized Floorplan
X Macros very close	✓ Macros well spaced
X Narrow routing channels	✓ Wide routing channels
X No proper power stripes	✓ Power stripes placed
X Congested (red blocks)	✓ Low congestion (blue)
X Long routes, high IR drop	✓ Short routes, low IR drop
X Timing & DRC issues	✓ Better timing & QoR

OPTIMIZATION CHECKLIST

- ☑ Die size & aspect ratio optimized
- ☑ Macros placed to reduce congestion
- ☑ Sufficient routing channels provided
- ☑ Power ring on all sides
- ☑ Power stripes & vias planned
- ☑ IO & pins placed with routing plan
- ☑ Blockages & keep-outs defined
- ☑ Congestion map within limits
- ☑ IR drop within spec
- ☑ Timing estimation OK
- ☑ DRC / Antenna preliminary clean
- ☑ QoR metrics within target

BENEFITS OF OPTIMIZED FLOORPLAN

- ✓ Low congestion & smooth routing
- ✓ Better timing & less skew
- ✓ Low IR drop & good reliability
- ✓ Less power & area
- ✓ Fewer ECOs & cost saving
- ✓ High overall QoR

PRACTICAL OBSERVATIONS

- * Start optimization early in floorplan.
- * Always check global routing & IR drop reports.
- * Reserve space for CTS, decap, filler, tap cells.
- * Balance area, timing, power and routability.
- * Floorplan optimization is iterative process.

SUMMARY POINTS

- Floorplan optimization refines the physical plan for best QoR.
- It involves macro placement, power grid, IO, routing resources and analysis.
- Proper optimization avoids congestion, IR drop, timing failures and costly ECOs.
- Iterate until all metrics meet the design goals and sign-off requirements.

AUTHOR : Konda Tharun Vivekananda

PD FLOOR PLANNING

12. FLOORPLANNING – ITS PROBLEMS AND THEIR SOLUTIONS

* INTRODUCTION

- Floorplanning defines the physical organization of the chip such as die size, macro placement, IO, power rings, routing channels, special cells, blockages etc.
- If floorplan is poor, it leads to congestion, timing violations, high power, low yield and costly ECOs.

* WHY FLOORPLANNING PROBLEMS OCCUR ?

- Incorrect die size / aspect ratio
- Poor macro / IO placement.
- Insufficient routing channels
- Improper power grid planning
- Ignoring congestion and timing criticality.
- Not following DRC / foundry rules

* MAJOR FLOORPLANNING PROBLEMS AND SOLUTIONS

S.NO.	PROBLEM	DESCRIPTION / EFFECT	SOLUTION
1.	CONGESTION	• Too many nets in limited routing space. • Detours, long wires, routing failures • Timing degradation.	• Provide sufficient routing channels • Increase core size or optimize aspect ratio. • Use congestion analysis early.
2.	IR DROP / POWER GRID ISSUES	• Uneven power distribution. • Voltage drop, timing failures, functional failures.	• Plan power rings and stripes. • More vias & wider metal. • Check IR drop early.
3.	TIMING VIOLATIONS	• Long critical paths. • High skew, high latency. • Setup / Hold failures.	• Place macros near critical logic. • Proper CTS planning. • Buffer insertion & shielding.
4.	AREA WASTAGE	• Unbalanced placement. • Excess white space, low utilization.	• Optimize macro placement. • Adjust core size. • Improve aspect ratio.
5.	DRC / LVS / ANTENNA RISKS	• Violations due to poor planning. • Rework and delays.	• Follow design rules early. • Reserve space for antenna, decap, filler.
6.	ECO / FUTURE SCALABILITY	• Poor plan → difficult ECOs. • No space for spare cells / fillers.	• Keep placement flexibility. • Reserve spaces. • Maintain symmetry.

* KEY POINT *

A good floorplan minimizes congestion, IR drop, timing issues and area wastage, improves QoR and reduces ECO cost.

PRACTICAL OBSERVATIONS

* Start with proper die size and aspect ratio.
* Macros near related logic improve timing.
* Always provide enough routing & power resources early.
* Check congestion and IR drop before placement.
* Floorplan optimization is iterative until all targets met.

VIVA QUESTIONS

Q1. Why is floorplaning important ?
→ It defines physical organization and decides QoR.

Q2. What happens if routing channels are insufficient ?
→ Congestion, detours, high delay and routing failures.

Q3. How do you solve IR drop in floorplan stage ?
→ Proper power rings, stripes, more vias and early IR analysis.

Q4. Why should macros be placed close to logic ?
→ To reduce long wires and improve timing.

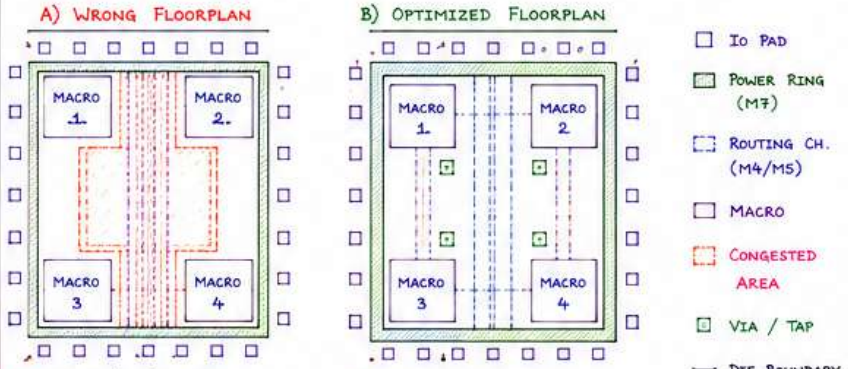
FLOORPLANNING FLOW WITH OPTIMIZATION (ITERATIVE)



STAGE BY STAGE OPTIMIZATION DETAILS

1) Analyze Design : Understand design size, I/O count, power, timing critical paths, routing demand and technology rules.
2) Initial Floorplan : Decide die size, aspect ratio, core area, place macros and I/Os roughly. Reserve space for power, clock, decap, fillers.
3) Congestion & Power Planning : Estimate routing demand, plan channels horizontally & vertically, insert power rings and stripes, add vias.
4) Timing Aware Optimization : Move macros near critical logic, adjust channels, keep outs, balance placement regions to improve timing.
5) DRC / Antenna Check : Verify spacing, metal rules, antenna ratios and fix issues early in floorplan.
6) Final Floorplan : Freeze plan after meeting PPA, QoR, rules and sign-off targets. Export DEF for placement.

EXAMPLES – WRONG FLOORPLAN vs OPTIMIZED FLOORPLAN



✗ Macros far from logic
✗ Narrow / missing channels
✗ Congested center region
✗ Poor power distribution
✗ Long wires, high delay

✓ Macros well placed
✓ Sufficient routing channels
✓ Proper power rings & stripes
✓ Balanced placement regions
✓ Short routes, better QoR

COMMON FLOORPLANNING PROBLEMS SUMMARY

1. Contributes to Congestion	2. Contributes to Timing Issues
• Overcrowded macros	• Long critical paths
• Narrow channels	• High clock latency
• Ignoring routing demand	• Low power integrity
• Poor aspect ratio	• IR drop → timing fail

CHECKLIST

- ✓ Correct die size & AR
- ✓ Macros balanced
- ✓ Enough routing channels
- ✓ Power rings & stripes
- ✓ DRC / Antenna clean
- ✓ IR drop within limit
- ✓ Timing Est. OK
- ✓ Ready for placement

BENEFITS OF GOOD FLOORPLANNING

✓ Low congestion	✓ High area utilization
✓ Less IR drop	✓ Less ECO cost
✓ Better timing	✓ Faster time to market
✓ Low power consumption	✓ Better overall QoR

SUMMARY POINTS

- Floorplanning problems create major issues in later stages.
- Early optimization saves area, power, timing and cost.
- Always iterate floorplan until all metrics meet targets.
- Good floorplan = smooth flow + high QoR + reliable chip.

PD FLOOR PLANNING

13. Important Terms of Floorplanning

1) Standard Cell Area
- Region where standard cells will be placed.
- Also called core / std. cell region.
- Occupies most of the core area.

2) Macro Area
- Large pre-designed blocks (memories, IPs, PLL, ADC, etc.).
- Placed in floorplan before std. cell placement.
- Need halos (keepout) around them.

3) Blockages Area
- Areas where placement or routing is restricted.
- Types: Hard, Soft, Partial, Routing blockages.
- Used to protect structures, fix congestion, obey rules.

4) Die Boundary
- Outer most boundary of the chip.
- Defines the total silicon area of the design.
- Includes scribe lines, seal ring, IO ring.

5) IO Ports - Input / Output
- Points on die boundary for input / output signals.
- Placed around periphery (IO ring).
- Includes data, control, status, test, analog I/O.

6) Clock Inputs
- Clock pins bring reference clock into the chip.
- Must be placed at ideal locations with low skew.
- Multiple clock domains → separate clock pins.

7) Power Rings (VDD / VSS Ring)
- Metal rings around the core to distribute power.
- Provides low resistance path to core & IO cells.
- Usually placed on top metal layers.

8) Power Stripes (Rails)
- Horizontal / vertical metal lines inside core.
- Connect power ring to std. cell rows.
- Control IR drop & ensure uniform power.

9) Channels
- Empty spaces reserved for routing.
- Types: Horizontal channel (between rows)
  Vertical channel (between macros / blocks)
  channel → easier routing.

10) Keepout / Halo
- Buffer region around macros / IPs / critical nets.
- No cells / no routing inside this area.
- Prevents coupling, crosstalk and timing issues.

11) Special Cells Area
- Location reserved for special cells like decap, level-shifter, tie-cells, well-tap, antenna diodes. Helps in power integrity & reliability.

12) Filler Cells Area
- Small cells inserted after placement to fill gaps.
- Improves density, timing & routing stability.

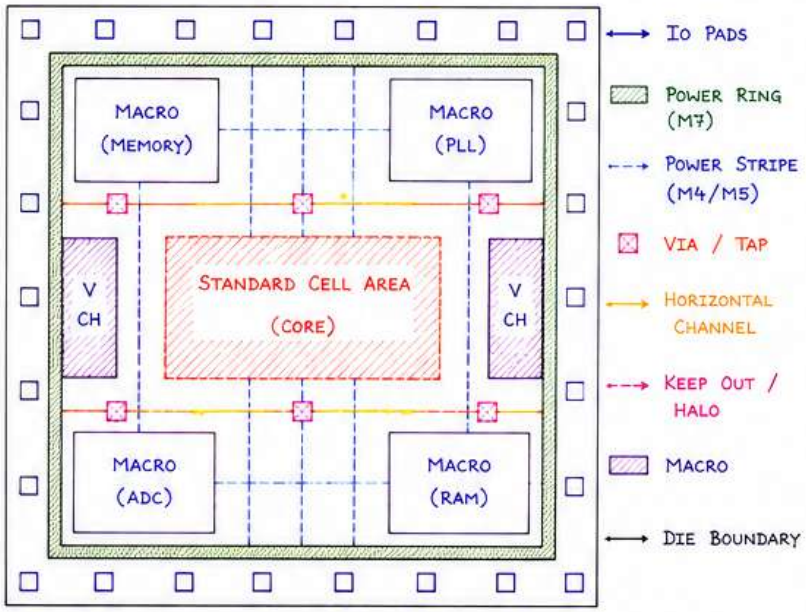
13) Tap Cells / Well Taps
- Placed in rows to connect wells to VDD / VSS.
- Avoid latch-up & improve substrate integrity.

*** Key Point ***

All these terms together decides the quality of the floorplan and directly impacts PPA (Power, Performance, Area) and overall QaR.

Viva Questions
Q1. What is standard cell area ?
→ Region used for placement of std. cells.
Q2. Why we need keepout around macros ?
→ To avoid DRC, coupling and routing blockage.
Q3. What is the purpose of power ring & stripes ?
→ To distribute power uniformly and reduce IR drop.
Q4. Difference between IO port and clock pin ?
→ IO port carries data/control; clock pin brings reference clock for timing.

Floorplan - Key Elements (Example)



Common Floorplanning Terms - Summary Table

S.No.	Term	Description	Purpose / Use	Location
1	Standard Cell Area	Area used to place standard cells.	Implements logic functionality.	Inside core region.
2.	Macro Area	Pre-designed large blocks (mem, IPs).	Provide complex functions.	Inside core (fixed loc).
3	Blockages Area	Restricted areas for placement / routing.	Protect structures, control congestion.	Inside core.
4.	Die Boundary	Outer boundary of the chip.	Defines total silicon area.	Outermost boundary.
5	IO Ports	Pins for input / output signals.	Interface with outside world.	Around die periphery.
6.	Clock Inputs	Pins for reference clock signals.	Synchronize sequential logic.	Periphery (preferred sides).
7	Power Ring	Metal ring around core.	Distributes power to core.	Around core region.
8	Power Stripes	Horizontal / vertical metal lines.	Reduce IR drop, connect rows.	Inside core.
9	Channels	Empty routing spaces.	Provide routing resources.	Between rows / macros.
10	Keep Out / Halo	Buffer around macros / critical nets.	Avoid DRC, crosstalk, timing issues.	Around macros, critical nets.
11	Special Cells Area	Space for decap, tie, tap, LS, etc.	Reliability, power integrity.	Inside core.
12	Filler Cells	Small cells to fill gaps.	Improve density, timing & routing.	Inside rows.
13	Tap Cells / Well Taps	Connect well/substrate to VDD/VSS.	Prevent latch-up, substrate integrity.	Inside rows. (periodic).

Why These Terms Are Important ?
- ✓ Proper allocation → better routing.
- ✓ Low IR drop & good power integrity.
- ✓ Less congestion & fewer ECOs.
- ✓ Better timing closure.
- ✓ Meets DRC / LVS / Antenna rules.
- ✓ Improves overall QaR and sign-off.

Practical Observations
- ✱ Start with die size and aspect ratio.
- ✱ Place macros early and provide halos.
- ✱ Reserve good space for power ring & stripes.
- ✱ Wide channels where routing demand is high.
- ✱ Keep clock pin near global clock source.
- ✱ Verify all rules before placement.

Benefits of Well Defined Floorplan
- ✓ Efficient area utilization
- ✓ Lower power & IR drop
- ✓ Reduced congestion
- ✓ Reliable routing & timing
- ✓ Less rework and ECOs
- ✓ Easy physical verification
- ✓ Better overall QaR
- ✓ Smooth path to sign-off

14. Flat Fullchip Floorplan and Hierarchical Fullchip Floorplan

* Introduction

- Floorplanning can be done in different ways.
- Two common approaches are :
  - → Flat (Non-Hierarchical) Fullchip Floorplan
  - → Hierarchical Fullchip Floorplan
- Choice depends on design size, complexity, team structure, reuse and QoR targets.

1) Flat (Non-Hierarchical) Fullchip Floorplan

- Entire chip is planned as one block.
- All macros, IOs, power grid, channels designed in a single level.

2) Hierarchical Fullchip Floorplan

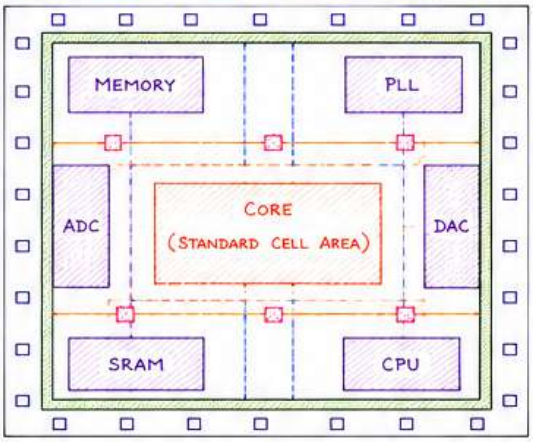
- Chip is divided into multiple blocks / sub-blocks (hierarchy).
- Each block is floorplanned separately & then integrated at top level.

*** Key Point ***

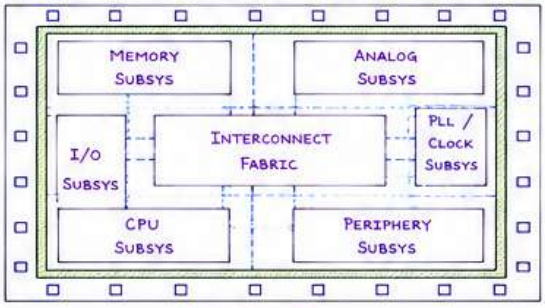
Flat floorplan is simple but not scalable. Hierarchical floorplan is scalable and reusable for large SoCs.

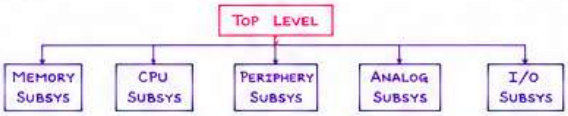
Examples

A) Flat Fullchip Floorplan (Example)



B) Hierarchical Fullchip Floorplan (Example)





Each sub-block is floorplanned in its own level and then integrated.

Viva Questions

Q1. What is flat fullchip floorplan ?
→ Entire chip is planned as a single level without hierarchy.

Q2. What is hierarchical floorplan ?
→ Chip is divided into sub-blocks and each block is floorplanned separately.

Q3. Which is better for large SoCs ?
→ Hierarchical floorplan is better.

Q4. Why do we need floorplanning ?
→ To decide physical organization for good QoR, timing, power and area.

IO PAD | MACRO | → VIA / TAP
POWER RING (M7) | → HORIZONTAL CHANNEL | ↕ VERTICAL CHANNEL
←→ POWER STRIPE (M4/M5) | → DIE BOUNDARY

Advantages – Flat Floorplan

- ✓ Simple to create and understand.
- ✓ No hierarchy overhead.
- ✓ Good for small to medium designs.
- ✓ Easier communication (single team view).
- ✓ Direct control of all resources.

Advantages – Hierarchical Floorplan

- ✓ Scalable for large designs (SoCs).
- ✓ Reuse of blocks / IPs.
- ✓ Easier team partitioning.
- ✓ Faster iterations (block level only).
- ✓ Good for ECOs at block level.
- ✓ Better timing closure (localized view).

Comparison – Flat vs Hierarchical Floorplan

S.No.	Parameter	Flat Floorplan	Hierarchical Floorplan
1.	Structure	Single level (one block)	Multiple levels (blocks / sub-blocks)
2.	Scalability	Poor for very large designs	Excellent for large SoCs
3.	Reusability	Low (everything from scratch)	High (reuse of blocks / IPs)
4.	Team Work	Hard (single team)	Easy (multiple teams)
5.	Iteration Time	Long (entire chip every time)	Short (only block level)
6.	Memory Usage	High (whole chip in memory)	Low (block by block)
7.	Timing Closure	Harder (global paths)	Easier (localized paths)
8.	ECO Handling	Time consuming	Faster (block level ECO)
9.	Design Examples	Small CPUs, small ASICs	Large SoCs, MPSoCs, Complex designs
10.	Initial Effort	Less	More (planning hierarchy)

Common Floorplanning Problems and Solutions

Problem	Solution
✗ Incorrect die size / aspect ratio	✓ Estimate early & iterate die size.
✗ Poor macro placement	✓ Place macros near related logic & balance.
✗ Congestion in routing channels	✓ Reserve wider channels & do congestion check.
✗ IR drop / EM issues	✓ Add more power stripes, rings & vias.
✗ Timing violations due to long paths	✓ Optimize macro locations & reduce wirelength.
✗ Insufficient power grid	✓ Plan complete power ring & stripe network.
✗ DRC / Antenna violations	✓ Follow design rules & antenna rules.
✗ Poor IO placement	✓ Group IOs by interface & direction.
✗ Low area utilization	✓ Remove dead space, optimize aspect ratio.

When to Use Flat Floorplan ?

- Small to medium size designs.
- Simple architectures.
- Limited reuse of IPs.
- Less team / single team flow.
- Early prototyping.

When to Use Hierarchical Floorplan ?

- Large complex SoC designs.
- Many reusable IPs.
- Multiple teams / long projects.
- Need fast iterations.
- Better QoR and ECO efficiency.

*** Key Point ***

Floorplanning decides the skeleton of the chip. Good floorplan → better placement → better routing → better timing → better QoR.

Practical Observations

- * Always check congestion & IR drop before placement.
- * Reserve space for clock tree & power grid early.
- * Keep macros near related logic to reduce wirelength.
- * Floorplan is iterative until all sign-off targets met.

Summary Points

- Floorplanning can be done as flat or hierarchical.
- Flat floorplan is simple but not scalable.
- Hierarchical floorplan improves scalability and reuse.
- Proper floorplan reduces congestion, IR drop and timing issues.
- Good floorplan leads to smooth physical design flow.

PD FLOOR PLANNING

15. Multi Voltage Domain Floorplanning

✱ Introduction

- Modern SoCs use multiple voltage domains (MVD) to optimize power, performance and functionality.
- High performance blocks use higher Vdd.
- Low activity blocks use lower Vdd.
- Each voltage domain is isolated and connected using level shifters where required.

✱ Why We Need Multi Voltage Domain ?

- Reduce dynamic power ($P \propto CV^2 f$).
- Operate blocks at the minimum required voltage.
- Improve energy efficiency and battery life.
- Better thermal management & reliability.
- Different voltage levels for analog / digital blocks.

✱ Power Saving Terms

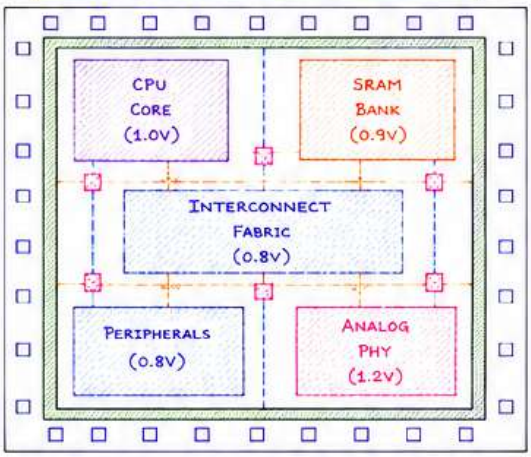
- DVFS (Dynamic Voltage & Frequency Scaling).
- Power Gating (Turn off idle domains).
- Back Biasing (Reduce leakage in standby).
- Multi-Vdd operation → Lower overall power.

✱ Voltage Domain Partitioning

1. Identify blocks and their performance / power needs.
2. Group blocks with similar Vdd requirements.
3. Define domain boundaries.
4. Add isolation cells, level shifters and retention flops.
5. Prepare separate power rings/stripes for each domain.
6. Verify connectivity, IR drop and timing closure.

> ✱ **Key Point** ✱
>
> Multi voltage domain floorplanning balances performance and power by assigning proper voltage to each block and ensuring safe isolation & communication.

Example - 3 Voltage Domain Floorplan



Viva Questions

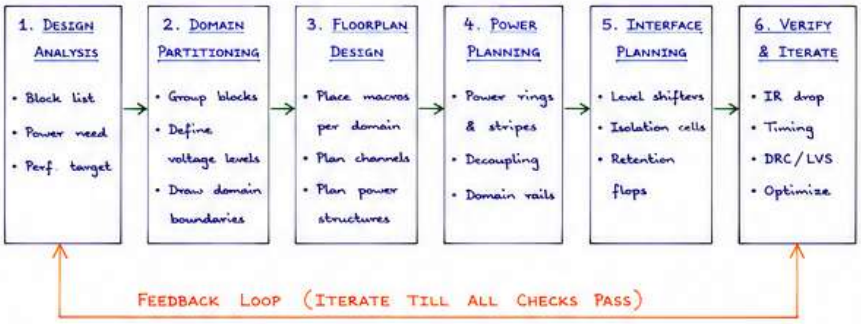
Q1. What is multi voltage domain design ?
→ Using multiple supply voltages in a chip to save power and improve performance.

Q2. Why level shifters are needed ?
→ To safely transfer signals between different voltage domains.

Q3. What is the role of isolation cells ?
→ To block leakage when a domain is OFF.

Q4. How MVD saves power ?
→ Low power blocks run at lower Vdd reducing dynamic power ($P \propto V^2$).

Multi Voltage Domain Floorplanning Flow



Advantages

- ✓ Significant power reduction.
- ✓ Better energy efficiency.
- ✓ High performance where needed.
- ✓ Lower leakage in low Vdd domains.
- ✓ Enables scalability for large SoCs.
- ✓ Thermal improvement.

Challenges

- ✗ Additional level shifters → area & power.
- ✗ Isolation & retention logic overhead.
- ✗ Complex power grid distribution.
- ✗ Timing closure across domains.
- ✗ Verification complexity increases.
- ✗ IR drop & noise management.

Flat vs Hierarchical (MVD Floorplanning)

Parameter	Flat Fullchip Floorplan	Hierarchical Fullchip Floorplan
Structure	Single level, entire chip in one plan	Chip divided into blocks / sub-blocks
Design Stage	Early stage or small designs	Mid to late stage / large SoCs
Scalability	Limited for very large designs	Highly scalable
Reuse	Low reuse	High reuse of blocks / IPs
Iteration Time	Long (entire chip each time)	Short (block level iterations)
Power Planning	Single global power grid	Distributed power per block
MVD Handling	Complex to manage	Easy domain management
Team Work	Difficult (single view)	Easy (multiple teams)
Verification	Heavy fullchip runs	Incremental block level
Best For	Small / medium chips	Large SoCs, complex chips

Common Interface Elements Between Domains

Element	Purpose	Placement
Level Shifter	Convert signal between different Vdd	At domain boundaries
Isolation Cell	Prevent leakage when domain is OFF	At outputs of OFF domain
Retention Flip-Flop	Hold state in low power mode	Inside sequential domain
Power Switch	Turn ON/OFF a domain	At domain power input

When to use Multi Voltage Domain ?

- When some blocks are performance critical.
- When power budget is tight.
- When different blocks have different Vdd tolerance.
- When low power standby is required.

Practical Observations

- ✱ Place high Vdd blocks near high current sources.
- ✱ Keep level shifters close to domain boundaries.
- ✱ Use wide power stripes for high Vdd domains.
- ✱ Balance IR drop across all domains.
- ✱ Early partitioning saves rework.

Benefits of Multi Voltage Domain Floorplan

- ✓ Lower dynamic & leakage power
- ✓ Better power management
- ✓ Supports DVFS & power gating
- ✓ Improved battery life
- ✓ Better performance per watt
- ✓ Enables heterogeneous blocks

Summary Points

- Multi voltage domain floorplanning assigns different Vdd to blocks based on their need.
- It is key technique for modern low power SoCs.
- Proper domain partitioning and interface planning ensures safe operation.
- Helps achieve power, performance and area goals.
- Verification and power planning are critical.

PD FLOOR PLANNING

16. Domain Placement Strategy

* Introduction

- Modern SoCs operate with multiple voltage domains (MVD) to reduce power and improve efficiency.
- Each domain runs at its own supply voltage (Vdd).
- Domains communicate through level shifters.
- Proper domain placement in floorplan is critical for power, timing, reliability and area.

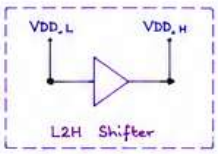
* Why We Need Level Shifters ?

- Different voltage domains have different logic thresholds.
- Direct connection → incorrect logic levels, failures.
- Level shifters safely transfer signals between domains.
- Prevent leakage, short-circuit current & timing issues. (Essential for reliable MVD operation)

* Types of Level Shifters

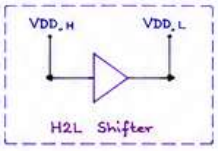
1) Low to High (L2H) Shifter

- Transfers signal from Low-Vdd domain to High-Vdd domain.
- Uses either dual supply design or charge pump.
- Unidirectional.



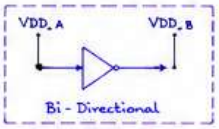
2) High to Low (H2L) Shifter

- Transfers signal from High-Vdd domain to Low-Vdd domain.
- Simple and area efficient.
- Unidirectional.



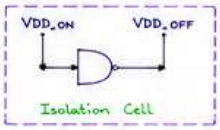
3) Bidirectional Level Shifter

- Allows signals to flow in both directions.
- Uses control signal or auto-direction scheme.



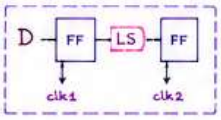
4) Isolation Cell (Not Exact Shifter)

- Used when domain is OFF.
- Prevents floating signals from propagating.



5) Data / Register Level Shifter

- Used in sequential paths.
- Includes retiming elements for safe clocking.



* Issues Without Level Shifters

- ✗ Incorrect logic levels → functional failures.
- ✗ Excess leakage current between domains.
- ✗ Short-circuit current & high power.
- ✗ Metastability and timing violations.
- ✗ Signal integrity degradation.
- ✗ Possible permanent damage to circuit.
- ✗ DRC / LVS violations (missing interfaces).

* Key Point *

In Multi Voltage Domain designs, level shifters are mandatory interfaces ensuring correct logic transfer, reliability and power efficiency.

Practical Observations

- * Place level shifters close to domain boundaries.
- * Avoid long low-Vdd wire driving high-Vdd loads.
- * Minimize number of crossings to reduce power.
- * Use proper power grid & decoupling for each domain.
- * Review placement with IR-drop & timing early.

Viva Questions

Q1. Why do we use level shifters in MVD designs ?
→ To safely transfer signals between different Vdd domains.

Q2. What are the types of level shifters ?
→ L2H, H2L, Bi-directional, Isolation cells, Register LS.

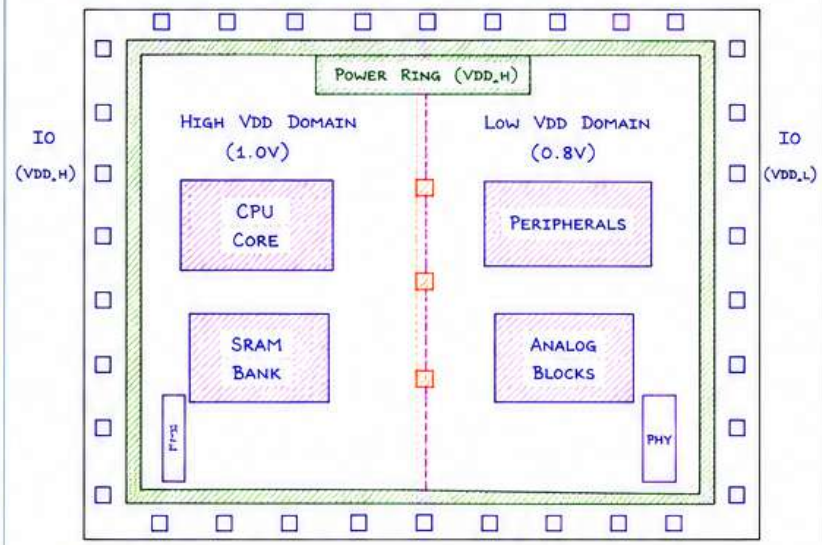
Q3. What happens if level shifters are not used ?
→ Logic failures, leakage, short-circuit, timing issues.

Q4. Where should level shifters be placed in floorplan ?
→ Near domain boundaries to minimize wire length.

* Domain Placement Strategy

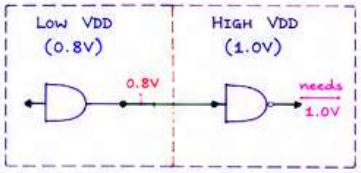
- Group blocks that communicate heavily in same domain.
- Cluster high performance blocks in high Vdd domain.
- Place off / low activity blocks in low Vdd domain.
- Keep level shifters at domain boundaries.
- Consider power grid, clocking and routing while placing.

* Floorplan Example (Two Voltage Domains)



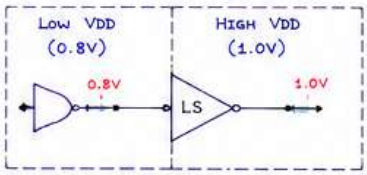
* Comparison : With vs Without Level Shifters

Without Level Shifter (L2H)



- ✗ Logic may not be recognized.
- ✗ High leakage / short-circuit.
- ✗ Timing violations.
- ✗ Functional failure risk.

With Level Shifter (L2H)



- ✓ Signal safely shifted to 1.0V.
- ✓ No leakage / short-circuit.
- ✓ Reliable operation.
- ✓ Meets timing & sign-off.

* Domain Placement Considerations

1. Communication	Minimize crossings between domains.
2. Performance	High speed paths in high Vdd domain.
3. Power	Place low activity blocks in low Vdd domain.
4. Power Grid	Separate grids for each domain, avoid IR drop.
5. Level Shifters	Place close to boundary & avoid unnecessary use.
6. Clocking	Use proper clock gating & domain specific clocks.
7. Verification	Check CDC, level shifters, isolation, retention.

Benefits of Proper Domain Placement

- ✓ Lower dynamic & leakage power.
- ✓ Better performance & timing.
- ✓ Efficient area utilization.
- ✓ Improved reliability.
- ✓ Easier power management.
- ✓ Better scalability for large SoCs.

Challenges

- ✗ More level shifters → area & power.
- ✗ Complex power grid distribution.
- * Increased verification effort.
- * Timing closure across domains.
- * Design and floorplan complexity.

Summary Points

- Multi voltage domain design is essential for modern low power SoCs.
- Domain placement strategy directly impacts power, performance and reliability.
- Level shifters are required at domain boundaries for safe signal transfer.
- Proper planning avoids leakage, timing issues and functional failures.
- Early domain planning and verification ensures smooth sign-off.

PD FLOOR PLANNING

17. Isolation Cells

* Introduction

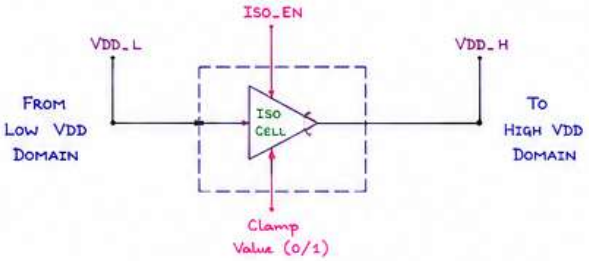
- Isolation cells are used at the boundaries between power domains to prevent invalid logic levels from propagating when a domain is turned OFF.
- When a domain is OFF, its outputs may float or go to unknown (X), which can corrupt logic in the always-ON domain.
- Isolation cell forces the output of the OFF domain to a known safe value (0 or 1).

* Why We Need Isolation Cells ?

- Prevent glitch / unknown values entering active domain.
- Avoid Functional failures in always-ON logic.
- Reduce leakage due to floating nets.
- Help achieve reliable power gating operation.
- Required for low power SoCs with power domains.

> *** Key Point ***
>
> Isolation cells protect the always-ON domain from unpredictable signals coming from a powered-down domain by clamping the output to a safe value.

* Basic Isolation Cell (Concept)



* Iso Cell Truth Table

ISO_EN	Low VDD Domain	ISO O/P (To High VDD Domain)	Interpretation
0	ON (Valid 0/1)	Same as input	Domain is ON, signal passes through.
0	OFF (Hi-Z / X)	0 (Clamp 0) or 1 (Clamp 1)	Domain OFF, output clamped to safe value.
1	ON (Valid 0/1)	0 (Clamp 0) or 1 (Clamp 1)	Force clamp value (used during power-down).
1	OFF (Hi-Z / X)	0 (Clamp 0) or 1 (Clamp 1)	Domain OFF, still clamped.

Note: Clamp value (0 or 1) is decided based on design requirement (usually 0 for active low signals).

* Types of Isolation Cells

1) Clamp 0 Isolation Cell
- When ISO_EN = 1 → output forced to '0'.
- Used for active-high data / control signals.

2) Clamp 1 Isolation Cell
- When ISO_EN = 1 → output forced to '1'.
- Used for active-low control signals.

3) Bi-directional Isolation Cell
- Used for buses / bidirectional signals.
- Isolates in both directions when ISO_EN = 1.

4) Multi-bit Isolation Cell
- Isolates multiple bits (bus) using a single cell.
- Saves area and power.

Viva Questions

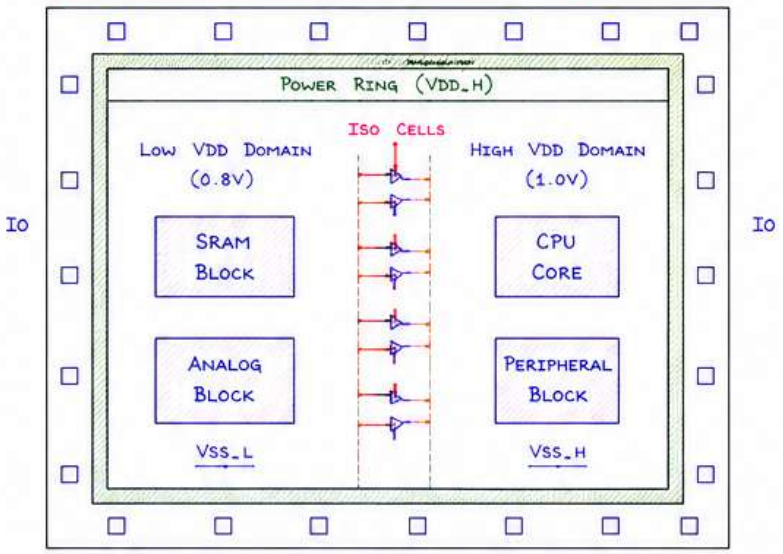
Q1. Why are isolation cells required in low power design ?
→ To prevent unknown values from entering the active domain when a domain is OFF.

Q2. What happens if isolation cells are not used ?
→ Functional errors, leakage, short circuit current, timing failures.

Q3. Where are isolation cells placed ?
→ At the output ports of the power domain, near domain boundary.

Q4. What is ISO_EN signal ?
→ Control signal that enables the clamp value during power-down.

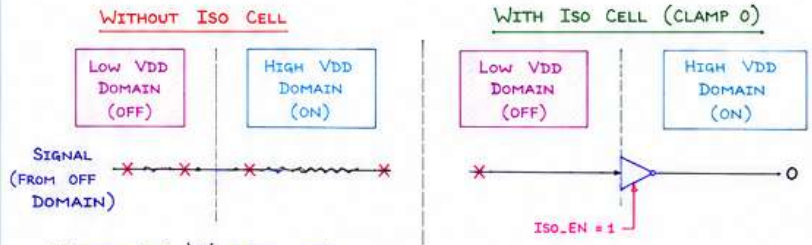
* Floorplan Example (With Isolation Cells)



- IO PAD
- POWER RING (M7)
- POWER STRIPE (M4/M5)
- HORIZONTAL CHANNEL
- ISOLATION CELL
- VIA / TAP
- DIE BOUNDARY
- DOMAIN BOUNDARY

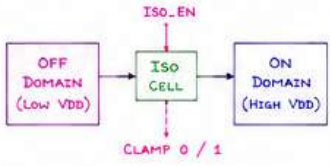
* Issues Without Isolation Cells

- ✗ Unknown / floating values propagate to always-ON domain.
- ✗ Large short-circuit current during switching.
- ✗ Functional failures in registers / state machines.
- ✗ Data corruption and system crash.
- ✗ Increased leakage due to floating nodes.
- ✗ Timing violations and unreliable operation.



* Placement Strategy

- Place ISO cells at all outputs of the OFF domain.
- Keep them close to domain boundary to reduce routing.
- Use multi-bit ISO cells on data buses.
- Ensure ISO_EN is controlled by power controller / FSM.
- Verify in STA and CDC to ensure correctness.



Benefits of Iso Cells

- ✓ Protects always-ON logic.
- ✓ Prevents leakage & short-circuit current.
- ✓ Increases design reliability.
- ✓ Supports deep sleep / power gating.
- ✓ Improves overall power efficiency.

Design Considerations

- Choose correct clamp value.
- Correct ISO_EN timing (before power-off).
- CTS / Reset network must be aware.
- Check for metastability on control.
- Verify during DRC / LVS / STA / CDC.

Summary Points

- Isolation cells clamp the outputs of OFF domain to a safe value.
- ISO_EN = 1 → output clamped (0 or 1), ISO_EN = 0 → normal pass.
- Types: L2H, H2L, Bi-directional, Isolation cell, Multi-bit Isolation cell.
- Without isolation cells → unknowns, leakage, short-circuit, failures.
- Proper placement at domain boundary is essential for power & reliability.

PD FLOOR PLANNING

18. Retention Flipflops

* Introduction

- Retention flipflops (RFFs) are special flops used in low power designs to retain state when a power domain is turned OFF.
- They save the state to a backup latch / memory and restore it when the domain turns ON.
- Essential for stateful elements in power gated domains.

* Why We Need Retention Flipflops ?

- To preserve sequential state during power shut-down.
- Avoid loss of context → faster wake-up.
- Reduces software / hardware re-initialization overhead.
- Improves performance and reliability.
- Critical for caches, FSMs, counters, registers, etc.

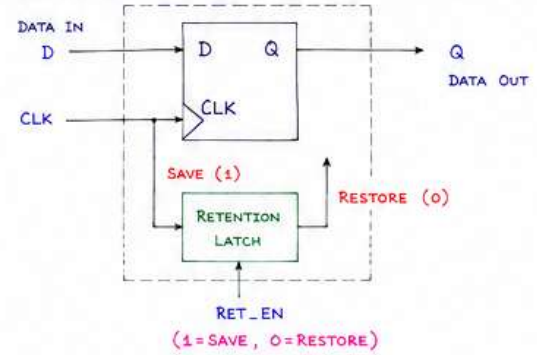
*** Key Point ***

RFFs store state when Vdd is OFF and restore the same state when Vdd is ON.
→ Minimum performance loss & better user experience.

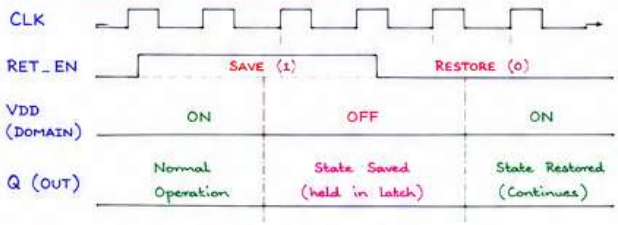
How Retention Flipflop Works ?

- When domain is ON: behaves like normal flipflop.
- When domain is going OFF (RET_EN = 1):
  - State is saved to retention latch.
- When domain is coming ON (RET_EN = 0):
  - State is restored from retention latch.

Retention Flipflop (RFF) - Conceptual Diagram



Retention Operation Timing



Issues Without Retention Flipflops

- ✗ State is lost when domain is power gated OFF.
- ✗ Need full re-initialization → slower wake-up.
- ✗ Wrong / incorrect state may lead to functional failures.
- ✗ Performace degradation due to software reload.
- ✗ Increased latency and power due to reconfiguration.
- ✗ State machines may enter invalid states.

Benefits of Retention Flipflops

- ✓ Preserves state across power OFF/ON.
- ✓ Faster wake-up time.
- ✓ Better performance and reliability.
- ✓ Reduces data loss and re-initialization.
- ✓ Essential for modern low power SoCs.

Viva Questions

Q1. Why are retention flipflops used ?
→ To preserve state of sequential elements during power shut-down.

Q2. What happens if retention flops are not used ?
→ State is lost, re-initialization is required, performance degrades.

Q3. What is RET_EN signal ?
→ Controls save/restore operation (1 = save, 0 = restore).

Q4. Where are retention flops used ?
→ In power gated domains for stateful elements like registers, FSMs, counters, caches, etc.

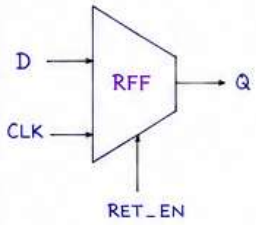
Retention Flipflop Truth Table

RET_EN (SAVE/RESTORE)	Domain (VDD)	RFF Output (Q)	Interpretation
0 (RESTORE)	ON	Q = Data from Retention Latch	Domain ON → State Restored
0 (RESTORE)	OFF	Previous State (Held)	Domain OFF but output is held in RFF
1 (SAVE)	ON	Normal Operation (Q follows D)	Normal flipflop mode
1 (SAVE)	OFF	State Saved in Retention Latch	Domain OFF → State stored safely

Types of Retention Flipflops

1) Scan Retention Flipflop
   - Retention + Scan capability.
2) Non-Scan Retention Flipflop
   - Only retention, no scan.
3) Multi-bit Retention Cell
   - Stores multiple bits to reduce area.
4) Retention Latch Based RFF
   - Uses latch as backup storage.
5) Memory Based Retention
   - Uses small SRAM for large number of flops.

Retention Cell Symbol



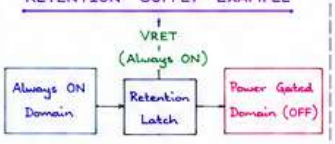
Standard Flipflop vs Retention Flipflop

Parameter	Standard Flipflop (DFF)	Retention Flipflop (RFF)
State During OFF	Lost	Saved in Retention Latch
Recovery / Wake-up	Needs Re-init (Slow)	Fast (State Restored)
Additional Hardware	No	Retention Latch + Mux
Control Signal	None	RET_EN (Save/Restore)
Area	Smaller	Slightly Larger
Power (Active)	Normal	Slightly Higher
Power (OFF)	Zero (lost)	Very Low (leakage)
Use Cases	Always-ON domains	Power Gated domains
Performance	Re-init delay	Better (fast wake-up)

Floorplan Considerations for RFFs

- Place RFFs at domain boundaries / close to flops.
- Ensure retention supply (VRET) is always ON.
- Use separate retention cell placement region if needed.
- Consider routing congestion for RET_EN and VRET.

Retention Supply Example



Practical Observations

- * RET_EN must be generated by power controller.
- * Save state before domain is completely OFF.
- * Restore should be done after domain is ON and stable.
- * Proper sequencing is crucial for safe operation.
- * Verify using STA, CDC and Power aware verification.

Summary Points

- Retention flipflops preserve the state during power shut-down.
- RET_EN controls save (1) and restore (0) operations.
- Without RFFs → state loss, re-init delay, functional issues.
- RFFs use retention latch/memory to store data safely.
- They improve performance, reliability and user experience.
- Proper placement, power and control planning is essential.

PD FLOOR PLANNING

19. AON (ALWAYS ON DOMAIN)

* INTRODUCTION

- AON domain (Always ON domain) is a special low power domain that remains powered even when other domains are shutdown.
- It contains essential control, retention, isolation, power gating logic and wake-up circuits.
- It acts as the brain of the chip during power-down.

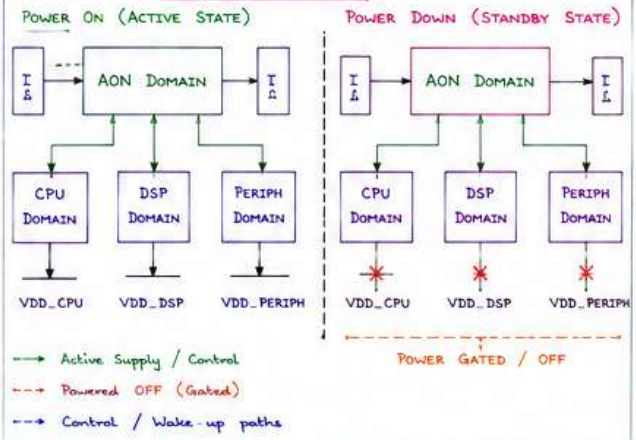
* WHY WE NEED AON DOMAIN ?

- To manage power gating (PG) of other domains.
- To control isolation cells, retention flops and level shifters.
- To generate wake-up events (external / internal).
- To store critical state and configuration.
- To ensure safe & controlled power down / up sequence.
- To maintain communication with external world.

* OPERATION OF AON DOMAIN

- AON domain is connected directly to primary supply (VDD_AON).
- Other domains (e.g., CPU, GPU, DSP) are power gated.
- AON keeps the system alive and monitors wake-up sigals.
- On wake-up, AON executes power-up sequence for other domains.

OPERATION DIAGRAM



* WHAT AON DOMAIN TYPICALLY CONTAINS ?

1. Power Management Controller (PMC)
2. Wake-up circuits (GPIO, RTC, Interrupt, Pattern detect)
3. Retention Flipflops & Retention Memory
4. Isolation cell enables and control logic
5. Level Shifter enables
6. Clocks / Oscillators (always on)
7. Configuration / strap registers
8. Power good monitors & reset generation
9. Scan chains / Debug logic

EXAMPLES OF WAKE-UP SOURCES

- External pin
- RTC alarm
- Interrupt
- Pattern match
- Watchdog timeout

* ISSUES WITHOUT AON DOMAIN

- ✗ No control logic available when domains are OFF.
- ✗ Unsafe power gating → inrush current / corruption.
- ✗ Loss of critical state → system may not resume correctly.
- ✗ Isolation & level shifter enables cannot be controlled.
- ✗ Wake-up events cannot be detected or generated.
- ✗ May lead to functional failures or chip hang.

*** KEY POINT ***

AON domain ensures that the chip remains intelligent, controlled and safe during power-down and enables reliable wake-up and operation.

VIVA QUESTIONS

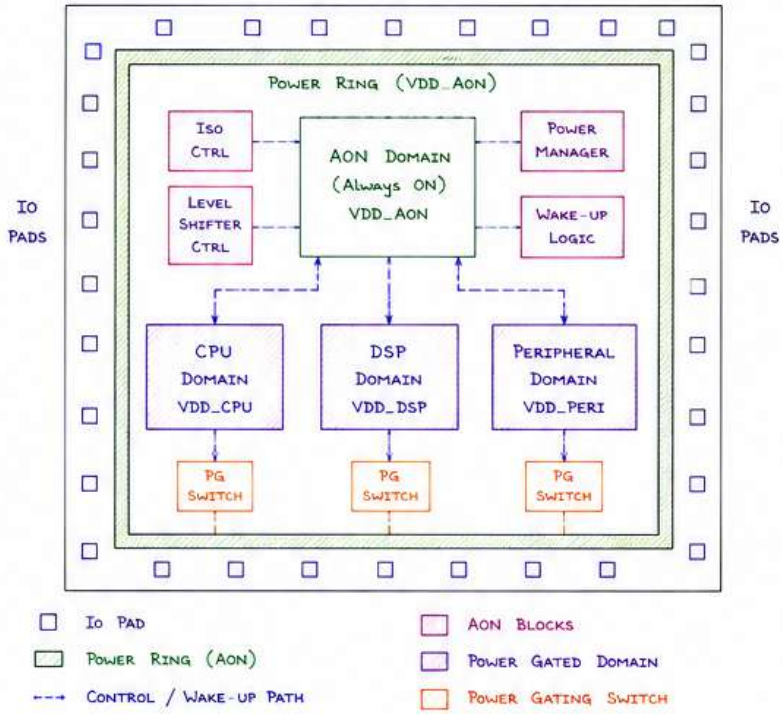
Q1. What is AON domain ?
→ A low power domain that remains always ON.

Q2. Why we need AON domain ?
→ To control power gating, isolation, retention and wake-up.

Q3. What happens if AON domain is not present ?
→ Loss of control, unsafe shutdown, state loss and system failure.

Q4. What kind of logic is placed in AON domain ?
→ PMC, wake-up logic, retention, isolation ctrl, clocks, config regs.

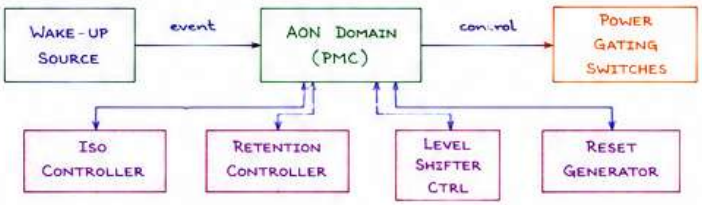
* AON DOMAIN FLOORPLAN EXAMPLE



* COMPARISON : WITH vs WITHOUT AON DOMAIN

PARAMETER	WITH AON DOMAIN	WITHOUT AON DOMAIN
Power Controll	Safe & controlled	Uncontrolled / unsafe
Wake-up Handling	Possible	Not possible
State Retention	Preserved	Lost
System Reliability	High	Low
Isolation / LS Control	Managed correctly	Cannot be managed
Power Sequencing	Proper sequence	Random / risky
Debug / Test	Always accessible	Not accessible
Overall Operation	Reliable & Robust	Unreliable & Error prone

* AON DOMAIN SIGNAL FLOW



* BENEFITS OF AON DOMAIN

- ✓ Reliable low power operation.
- ✓ Safe power-down and power-up.
- ✓ State retention for quick resume.
- ✓ Supports wake-up from deep sleep.
- ✓ Better performance & user experience.
- ✓ Essential for modern low power SoCs.

* PRACTICAL CONSIDERATIONS

- * Keep AON small to reduce static power.
- * Place AON near power ring for IR drop.
- * Ensure robust power supply (VDD_AON).
- * Use low leakage cells and well strategy.
- * Verify wake-up path, CDC and resets.
- * Consider routing congestion to AON.

SUMMARY POINTS

- AON domain is the always ON brain of the chip.
- It controls power gating, isolation, retention and level shifters.
- It detects wake-up events and sequences power-up.
- Without AON domain, system may lose state and fail.
- Proper placement and planning of AON is critical.
- AON ensures reliable, safe and efficient low power operation.

PD FLOOR PLANNING

20. Power Grid Separation & Power Switch

* Introduction

- In multi power domain SoCs, each domain has its own supply voltage and power grid.
- Power grid separation ensures isolation of IR drop, switching noise and leakage between domains.
- Power switch is used to connect / disconnect a domain from the main supply.
- It is one of the key aspects of floorplanning & power planning.

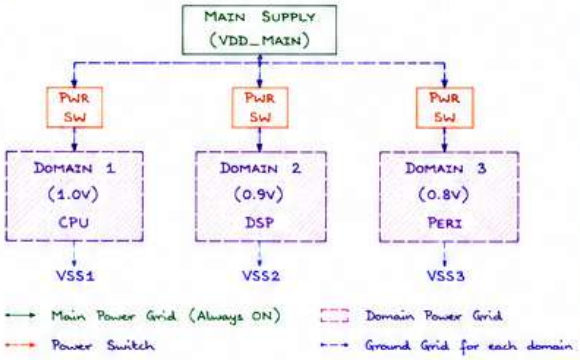
* Why We Need Power Grid Separation ?

- Reduce IR drop & voltage noise coupling between domains.
- Prevent cross domain leakage currents.
- Support different voltage levels for each domain.
- Improve power efficiency by turning OFF unused domains.
- Ensure reliable power delivery and safe operation.

> *** Key Point ***
>
> Power grid separation + Power switch control enables safe power gating, reduces leakage & improves power management in modern low power SoCs.

* Power Grid Separation Concept

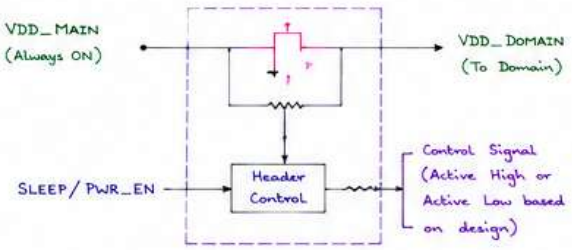
- Each domain has its own VDD / VSS network.
- Domains are isolated from each other except at the interface (level shifter, iso cells).
- Global power grid connects to each domain through power switches.



* Power Switch

- A high side switch placed between main VDD and domain VDD.
- Controlled by Power Management Controller (PMC).
- Cuts off power to the domain during power-down.
- Provides isolation from main supply.

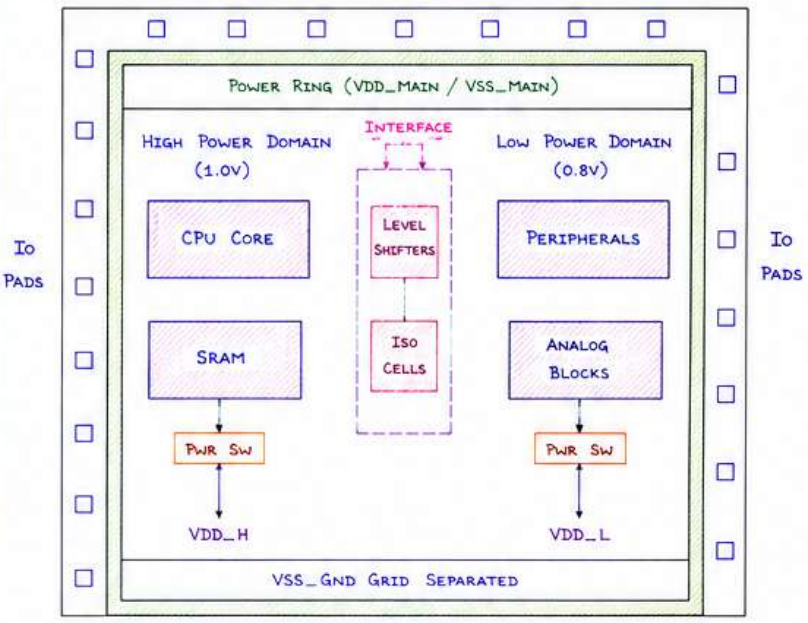
Power Switch Block (High Side PMOS Based)



* Benefits of Power Grid Separation & Power Switch

- ✓ Enables power gating for leakage reduction.
- ✓ Improves power integrity and reduces IR drop.
- ✓ Supports multi voltage operation.
- ✓ Isolates switching noise and cross talk.
- ✓ Improves performance and reliability.

* Power Grid Example (Two Power Domains)



* Comparison : With vs Without Power Grid Separation

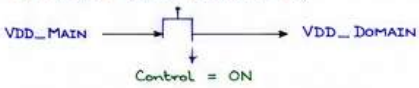
Parameter	With Separation	Without Separation
IR Drop	Localized & controlled	High coupling, large IR drop
Noise Coupling	Reduced	High noise interference
Leakage	Low (isolated)	High cross domain leakage
Voltage Flexibility	Different VDD levels possible	Single VDD only
Power Gating	Possible with power switches	Not possible
Reliability	High	Low
Area Overhead	Slightly higher (switches)	Lower but unsafe
Power Efficiency	High	Low

* Issues Without Power Grid Separation

- ✗ Large IR drop in one domain affects other domain.
- ✗ Switching noise propagates to sensitive analog blocks.
- ✗ Leakage currents flow between domains.
- ✗ Domain cannot be completely turned OFF.
- ✗ Timing violations due to voltage variation.
- ✗ Reliability and functional failures.

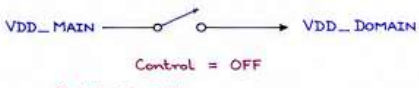
* Power Switch Operation

1) Active Mode (Domain ON)



- Switch is ON.
- Power flows to domain.
- Domain operates normally.

2) Sleep Mode (Domain OFF)



- Switch is OFF.
- Domain is disconnected.
- State held by retention flops.
- Leakage reduced.

* Types of Power Switches

1. Header Switch (High Side)
2. Footer Switch (Low Side)
3. Dual Switch (Header + Footer)
4. Integrated Switch Cells (Library based)

Viva Questions

Q1. What is AON domain ?
→ Always ON domain that remains powered during system sleep / power-down.

Q2. Why is power grid separation required ?
→ To reduce IR drop, noise coupling & leakage.

Q3. What is power switch used for ?
→ To turn ON / OFF a power domain.

Q4. What happens if power grid is not separated ?
→ Noise coupling, leakage, timing failures, high IR drop.

Summary Points

- AON domain remains always ON and is the backbone of low power SoCs.
- Power grid separation isolates domains and improves power integrity.
- Power switches control power flow to domains for power gating.
- Without separation, noise, leakage and IR drop cause functional failures.
- Proper floorplanning ensures safe, reliable and efficient power management.

21. Multi Voltage Verification

* Introduction

- Multi voltage SoCs have blocks operating at different voltage domains.
- Verification must ensure correct functionality across domain crossings (CDC), power states and level shifts.
- It checks logic, power intent, isolation, retention and timing across all voltage domains.
- Early verification avoids late re-spins and functional failures.

* Why We Need Multi Voltage Verification ?

- Ensure safe & correct data transfer between domains.
- Verify level shifters, isolation cells, retention flops.
- Detect power intent / UPF mistakes early.
- Ensure no floating / X propagation in power OFF modes.
- Guarantee timing closure across all voltage domains.
- Improve power, reliability and system robustness.

> *** Key Point ***
>
> Multi voltage verification ensures functionality, power intent correctness and timing safety across all voltage domains and power states.

* Verification Steps

1) **Collect Inputs**
   - Netlist (gate level)
   - UPF / CPF (power intent)
   - SDC (constraints)
   - Libraries (multi VT, multi voltage)
   - Floorplan / Voltage domain info
2) **Power Intent Check**
   - Run UPF/CPF check :
     - Syntax check
     - Connectivity check
     - Power state table validation
3) **CDC (Clock Domain Crossing) Verification**
   - Detect crossings between voltage domains.
   - Check synchronizers, protocol correctness.
   - Verify Async resets / handshake signals.
4) **Level Shifter Verification**
   - Check direction (Low->High / High->Low).
   - Verify placement near domain boundaries.
5) **Isolation Cell Verification**
   - Check ISO insertion at outputs of OFF domains.
   - Verify ISO control logic and clamp values.
6) **Retention Verification**
   - Check retention flop replacement.
   - Verify save/restore control & retention supply.
7) **Power Gating Verification**
   - Verify power switch insertion & control.
   - Check state sequence (OFF → ON wake-up).
8) **Functional Verification**
   - Run sims with power aware simulation.
   - Test all power states & voltage combinations.
   - Verify no X-propagation / unknown states.
9) **STA (Multi-Voltage Timing Analysis)**
   - Run multi-corner STA with voltage scaling.
   - Check timing across domains & LS delay.
10) **Sign-off Checks**
    - Power intent sign-off
    - CDC sign-off
    - Lint / DRC / LVS
    - IR drop / EM / SI

Viva Questions

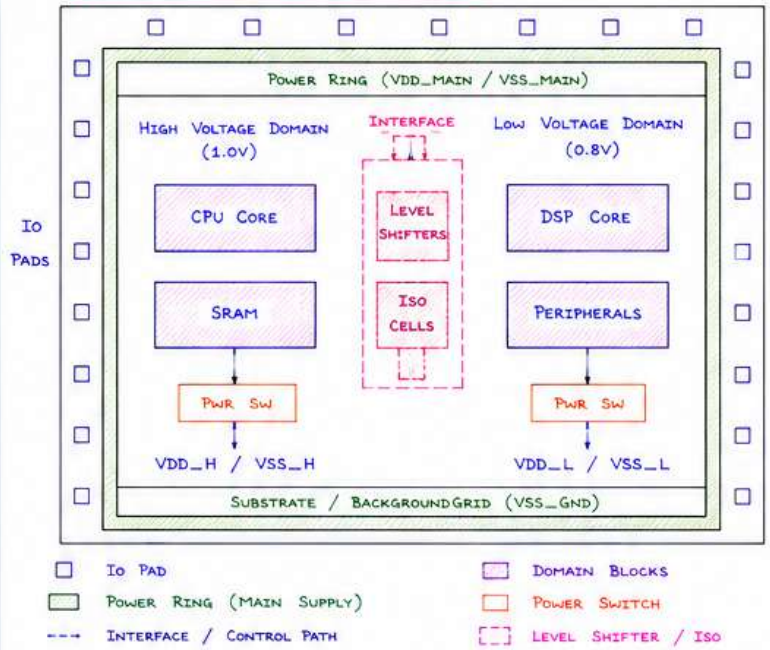
Q1. Why is multi voltage verification important ?
→ To ensure correct & reliable operation across voltage domains and power states.

Q2. What are the key elements verified ?
→ Level shifters, isolation cells, retention flops, power intent, CDC and timing.

Q3. What happens if LS is missing ?
→ Wrong logic levels / functional failures.

Q4. What is the role of isolation cells ?
→ Clamp outputs of OFF domains to safe value.

* Multi Voltage Domain Example



* Ways of Multi Voltage Verification

1) **Static Verification**
   - UPF/CPF check, Lint, CDC static, DRC/LVS.
2) **Dynamic (Simulation) Verification**
   - Power aware simulation with UPF.
   - Test all power states & voltage combinations.
3) **Formal Verification**
   - Formal CDC, Equivalence check across power states.
4) **Power Aware STA**
   - Multi-corner, multi-voltage timing analysis.
5) **In-silicon (Post Silicon) Verification**
   - Power state testing, scan based validation.
   - Built-in self test.

Power State Example

State	VDD_H (1.0V)	VDD_L (0.8V)
S0 (Active)	ON	ON
S1 (Low Domain OFF)	ON	OFF
S2 (High Domain OFF)	OFF	ON
S3 (Both off)	OFF	OFF

* Challenges in Multi Voltage Domain Floorplaning / Verification

Challenge	Description	Impact
IR Drop & Noise	Different voltage levels create IR drop & noise coupling between domains.	Timing failures, incorrect operation
Level Shifter Overhead	More LS required near domain boundaries.	Area ↑, Power ↑
CDC Complexity	High number of crossings between domains.	Functional & timing issues
Power Intent Errors	Wrong UPF/CPF leads to incorrect control.	Power up/down failures
Isolation Timing	Incorrect ISO control / clamp values.	Glitches, X-prop
Retention Management	Improper save/restore sequence.	State loss, data corruption
Verification Complexity	Many power states & voltage combos.	Long verification time
Debug Difficulty	Hard to find issues across multiple domains.	Late bug detection

* Issues Without Proper Verification

- ✗ Functional failures in power down state.
- ✗ Floating outputs causing X-propagation.
- ✗ Wrong level shifting → signal corruption.
- ✗ State loss in retention registers.
- ✗ High leakage & power waste.
- ✗ Timing violations across domains.
- ✗ System crash / unreliable operation.

* Benefits of Proper Verification

- ✓ Correct & reliable functionality.
- ✓ Safe power gating & wake-up.
- ✓ Reduced IR drop & noise.
- ✓ Optimized area & power.
- ✓ Robust timing across domains.
- ✓ Early bug detection & fix.
- ✓ Smooth tape-out & silicon success.

Practical Observation

- * Plan domains & boundaries early in floorplanning.
- * Place LS & ISO cells close to boundaries to reduce delay.
- * Verify power state sequence with power controller.
- * Always run power aware simulation before sign-off.

Summary Points

- Multi voltage verification ensures correct operation across domains.
- It verifies power intent, CDC, LS, ISO, retention & timing.
- Uses static, dynamic, formal, STA & in-silicon verification.
- Helps prevent late bugs and functional failures.
- Power states, isolation, retention & switches must be checked.
- Proper verification improves reliability, power & performance.
- Early planning & verification is key for modern low power SoCs.
- Tape-out quality depends on complete multi voltage verification.

RTL NETLIST → LIBRARIES → ELABORATION → SDC & CHECKS → MAPPING → OPTIMIZATION → SIGNOFF

FP PLAN

THANK YOU!

Your time and attention are greatly appreciated.

IN

OUT

Floor Planning – Transforming Architectural Intent into Optimized, Manufacturable, Signoff-Ready Layout.

FLOOR PLANNING OPTIMIZATION GOALS

Area Efficiency	Power Efficiency	Timing Closure	Congestion Reduction	Signal Integrity	DFM Compliance
m^2					

KEY ENGINES

TIMING ENGINE	CONGESTION ENGINE	POWER ENGINE	DRC / LVS ENGINE	QOR ENGINE
• STA • Slack Analysis • Path Optimization	• Hotspot Detection • Resource Balance • Re-Routing	• Power Estimation • IR Drop Analysis • Power Optimization	• Rule Checking • LVS Verification • Clean Signoff	• Area / Power / Timing • Multi-Corner Analysis • QoR Reporting

Thank you for being a part of this learning journey.
Keep Exploring, Keep Innovating, Keep Building the Future!

Author :
Konda Tharun Vivekananda

CLK

VDD

FLOORPLAN → PLACEMENT → CTS → 1 1 0 1 0 0 1 1 1 → SIGNOFF → GDSII